Home Network & Media Wall Plan

Revised Oct 20, 2026 · @emma · version 18 (revision 18: your plant stand and record shelf in two layouts, the PC's glass side, an edit mode in the 3D room), drawn to your measurements

What changed in v18: your plant stand and your record shelf, and a room you can rearrange. Your new plant stand (two columns, four tiers, grow lights under the top two) and **your 2 × 2 record shelf** (four trays) are in, with two layouts to choose from (section 4): **A, recommended**, the plant stand in the window's west corner, where it gets daylight as well as its grow lights, and the record shelf on the chimney breast under the “now playing” ledge, a step from the turntable; or **B**, the other way round. Your red lamp moves to the sofa's end either way. Both are drawn at assumed sizes until you measure them (two new questions). **The PC's clear side** now shows its graphics card, cooler, memory and fans. In the 3D room, **Edit room** lets you drag things around like a dolls' house, turn them, recolour them, take them out and add new ones (an armchair, a bookcase, a floor plant and more), and paint a wall, the ceiling or the skirting by tapping it; the menus are shorter for it. **HD with shadows** no longer asks more of the graphics card than it has, and if a phone does give up, the room comes back in Fast. More ideas for later (appendix). Fit proof 87 checks, none fail.

v17: the room finished round your notes. The sofa's open end is now a proper ottoman end: the seat wraps round to the back as one long cushion and the frame stops 20 short, and the sofa stands 13 cm off the window wall, as in your photo. **The window** has its real deep sill, 22 cm into the reveal, with your plants along it (3D room), and the **curtains** now hang clear of it. **Even light by default:** every spot head points straight down, evenly spread, and a new **General** scene comes first; tilt a head only if its pool lands badly. Your **red lamp's top (green) head** lights the wall with your shelf. **The flip-top** is drawn as built: pegged breadboard ends and three brass hinges. **The stairs:** one wall all the way up in one colour, the living room ceiling's edge where it cuts off the view, and the banister on the room side only, as you said. **The PC** is drawn with its insides, and lights up in the 3D room (the "PC lit" button); the plan adds a **USB hub** by the sofa, a **remote power button** and a **dust filter** (section 6), and checks the soundbar against the TV's remote (section 3). Your **OBEGRÄNSAD** (905.262.63) is confirmed at IKEA's 52 × 36 × 6. The build guide PDF now **stands on its own**: each step has its own cut list and pictures. New **ideas**, off the plan and the costs, are in their own tab and in the 3D room's buttons, all off until you turn them on (appendix). Fit proof 87 checks, none fail.

v16: the whole wall, redesigned round everything we now know. No dead space in the record bay: with LPs only, bay 3 splits in two. Under the turntable, two **snug LP crates** (a 33 cm trough, the width record shops use) still hold about 180 LPs face-out; beside them a **slim column** of three drawers: **two face-out game drawers** (about 44 DVD-size cases, flipped like the records) and a drawer for the **pads and remotes** with a charging lead, so nothing lies on the worktop (section 4). **The table by the stairs is now the default**, so the chimney breast is free: it takes a **“now playing” oak ledge** beside the turntable for the sleeve of what's on and the next two, under the two framed LPs. **Smart colour strips** in the upright grooves and under the side-wall run, in the same app as the bars' bulbs, so one scene sets the whole wall. On your shelf by the stairs: your **IKEA OBEGRÄNSAD**, black, its 16 × 16 white LEDs drawn showing a pixel music note; **the laser is there but off**; the **plant hangs from the shelf's window end**; and the **hollow pipe boxing** in the window corner (about 23 × 22, floor to ceiling) is drawn in, the shelf stopping just short of it. **Your hats** are on your red lamp, as in your photo: the graduation cap on top of the pole, the navy cap with its poem hung below the top head. **The sofa's end towards the TV is open**: no arm, one long seat cushion wrapping round to the back, an ottoman-style end. Fit proof 87 checks, none fail. Earlier rounds are in the appendix.

1 · The plan on a page

Drawn to your measurements, in Card Room Green: the alcove with the turntable zone on the right and bay 3 split into the slim game column and the LP crates; the chimney breast free, with its “now playing” ledge (the table is by the stairs, behind this view); both spot bars. Full-resolution renders come with this plan.

£3,110–6,337
the whole plan, item by item in section 11; no electrician on top

59 of 87
fit checks pass outright; 20 are tight, 8 need a check on site (section 3)

191 · 4–8
LPs, all face-out · seats at the table

Area	Decision
Alcove	The TV between two oak uprights, the left one now a corner post on the side wall; on the right, the turntable on the worktop and the Series X on a shelf at 134; below, three 60 cm bays: the Xbox consoles, the PlayStations with the sub, and (v16) two snug LP crates beside a slim column of game and pad drawers.
Side wall	One oak run from the corner: a 7 cm ledge with four more LPs that sweeps out into the PC's shelf above the sofa's open end, a smart colour strip underneath, clear of the TV's sightlines.
Dining	A flip-top gate-leg oak table that stays on the wall, now by default under your shelf by the stairs (v16): folded flat every day; open on two Häfele Hebgo brackets with its second half on a gate leg, 110 × 160, up to seven. The chimney breast keeps a “now playing” ledge and two framed LPs (option A puts the table there instead, up to six).
Records and games	Every LP face-out: about 191 (180 in the LP crates, 8 on the two ledges, 3 on the “now playing” ledge). Games: about 92 face-out or spine-up in the game drawers, and 32 PS1 games in the plinth drawer.
Heights	Worktop and table 76 · TV centre 116.5 · Series X shelf 134 · ledges and PC shelf 155.4 · high shelf 196, 39 under the 235 ceiling.
Colour and light	Card Room Green No. 79, colour-drenched. The ceiling LED strip, smart colour strips in the uprights and under the side run, a bias light, and two Wolston-style spot bars with smart colour bulbs on your two existing ceiling points; one app, one scene for the lot.
Power and network	Your own job: the side-wall socket becomes a double, left for the 10-gang (everything modern, the PC included), right for the retro set, switched at the wall. Deco A in bedroom 3, Cat6 down the airing cupboard.

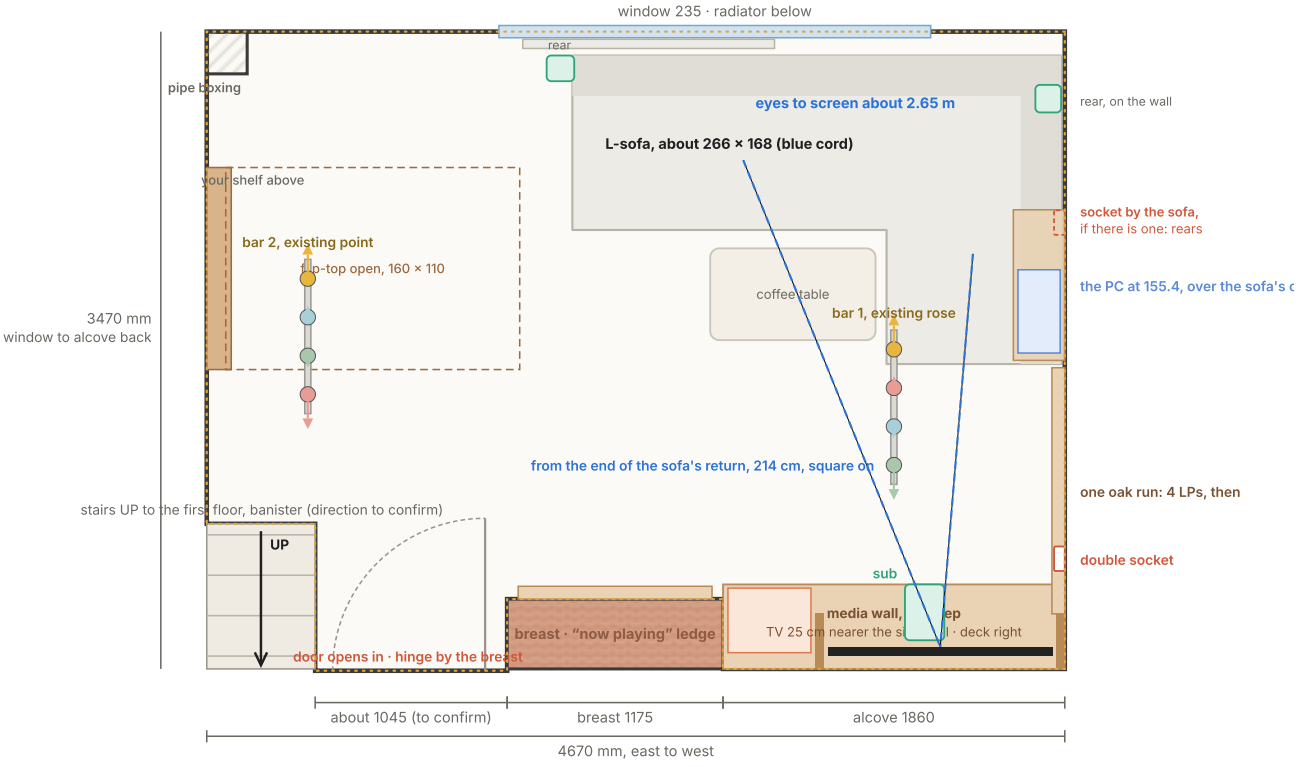
To confirm (section 12): the stairs wall's build · that you're swapping the light fittings yourselves · the sofa's return length · the plant stand's size · the record shelf's size · the shelf's height · the turntable's model · the fan stand's model · chairs · record counts · game counts · extra wired ports · the 10-gang · console model numbers · the pendant point · the RetroTINK's size.

2 · The room

Everything is drawn from your hand-measured sketch; where the agent's floor plan disagrees, your tape wins. The L-sofa is drawn from your photo[24]: its long side ends in line with the breast's reveal (about 266 cm); the return along the side wall is a typical 168 until you measure it. Since v17 its end towards the TV is an **ottoman end** (no arm; the seat wraps round, the back stops 20 short) and it stands 13 cm off the window wall, clear of the curtains. The window has a **deep sill**, 22 cm into the reveal at 88 from the floor, with your plants along it.

The living room from above, as you measured it

To scale, north at the top. Media wall in the alcove, one oak run on the side wall over the sofa, the flip-top table by the stairs (the default), two spot bars



Ceiling 2350 mm. Sofa from your notes (long side to 40 cm short of the window's end; return 168 assumed). Both spot bars run from the TV towards the window.

The whole room in 3D, from the window on an everyday evening, the table folded by the stairs: your blue sofa with its open end, the rug, the bars running towards the window, the “now playing” ledge on the breast. On the plan page you can walk round it, switch the table's wall, slide every light, colour the smart bulbs, paint the walls, ceiling and skirting separately, and save any view in HD.

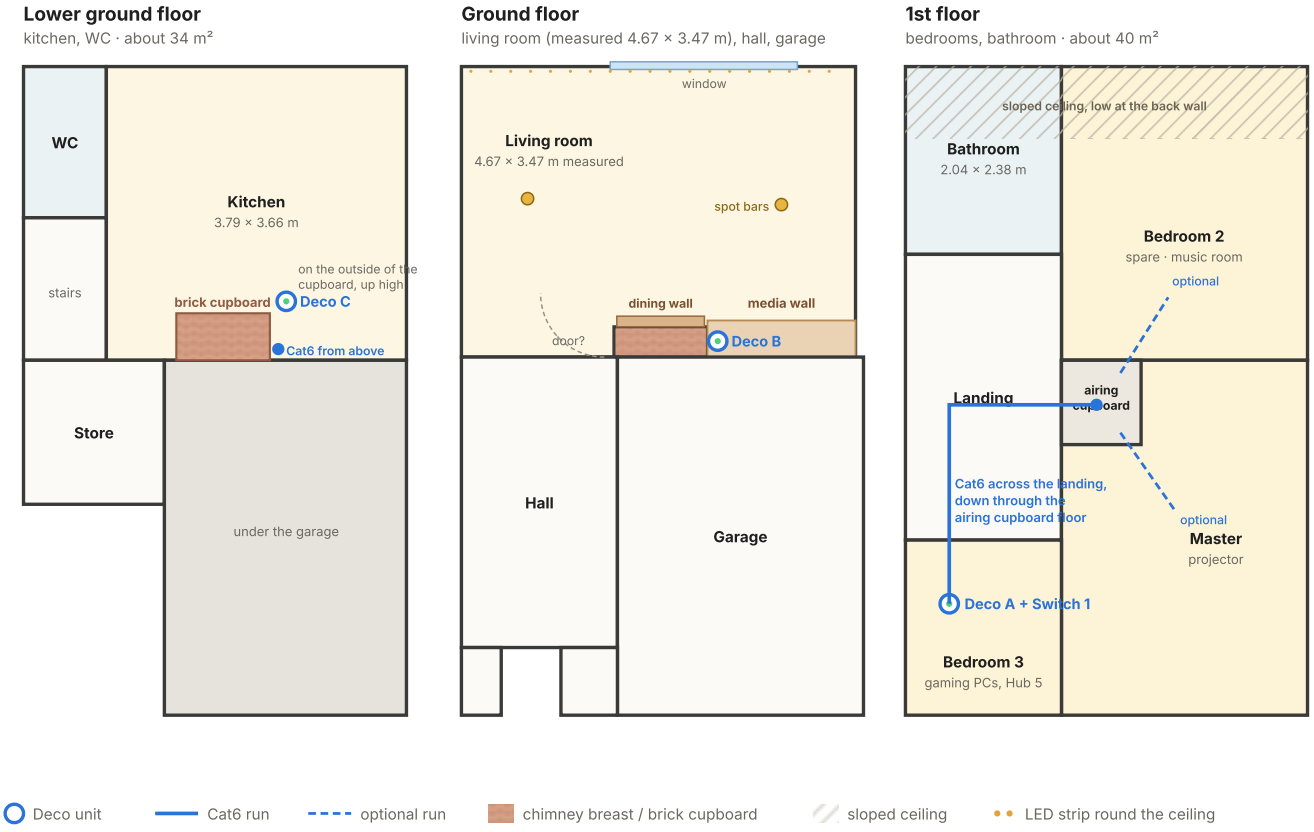
From your sketch	mm	Notes
Room, east to west along the window	4670	The media wall is the south wall, shared with the hall and garage
Window wall to the alcove's back wall	3470	To the chimney breast's face: 3090
Ceiling	2350	
TV alcove, east wall to the breast	1860 × 380	Floor to ceiling
Chimney breast	1175	Projects 380 on the alcove side, 390 on the west side: the dining wall
Window and radiator (north wall)	2350 · 1370	Window starts 730 from the east wall; radiator ends 130 inside its west end; the glass from 880 to 2120 up, set 220 back in its reveal over a deep sill (v17, from your note; the depth is drawn, please check it)
Skirting	60 × 20	White now; Card Room Green eggshell in the plan
West end: recess, door, stair box	935 / 1045 / 660 / 820	Drawn roughly: please confirm which is which
Stairs, beside the door	820 wide	They go **up** to the first floor (you confirmed), behind a wooden banister; drawn rising away from the room until you send a photo

From the sofa: from the middle of the long side your eyes are about 2.65 m from the screen, which a 55-inch fills at about 26°: right for TV and games. With the TV 25.25 cm nearer the side wall, it now sits square on to the end of the sofa's return (214 cm away), and only about 14° off-centre from the middle of the long side. The screen's centre at 116.5 cm is 1–2.5° above a seated eye line. Both spot bars hang 22.4 cm, so the lowest shade is at 212.6 cm: no headroom worries.

The house

The house: where the Decos and Cat6 runs go

Deco A and Switch 1 move to bedroom 3 with the Hub 5. Both Cat6 runs drop through the airing cupboard, right over the media wall's corner



Outline traced from the agent's plan (north, the window side, at the top). The living room's alcove, chimney breast and windows are placed from your measurements.

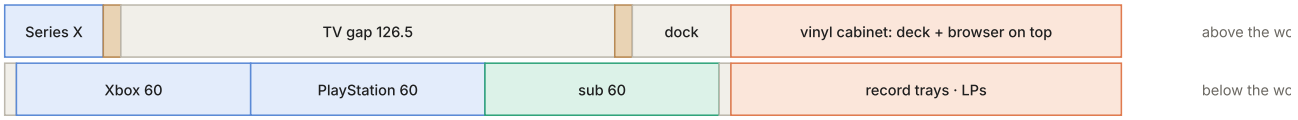
Deco A and Switch 1 sit with the Hub 5 and the gaming PCs in bedroom 3; both Cat6 runs go across the landing and drop through the airing cupboard, which sits over the alcove's corner (section 9).

3 · The media wall

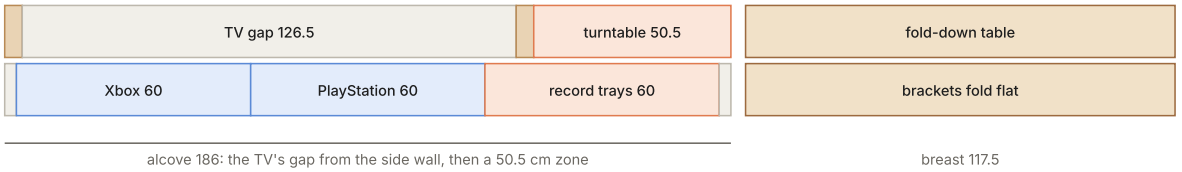
What moved in v11

To scale, from the side wall. Top: v10. Bottom: v11

v10: the TV in the middle, the vinyl cabinet on the breast



v11: the TV 25.25 cm nearer the side wall, the dining wall on the breast



- Moving the TV 25.25 cm left joins the two 25.25 cm side gaps into one 50.5 cm zone: wide enough for a 45.2 cm turntable, with 2.65 cm each side.
- The Series X moves up onto a shelf over the deck, above the lid's swing, with 15 cm of air each side. The records sit in bay 3, right under the deck.
- That frees the chimney breast for a fold-down dining table; the sub, which is wireless, moves to the floor by the side wall.

Why this layout, and what we ruled out

- **The TV in the middle (v10):** left two 25.25 cm gaps either side, too narrow for a turntable, so the vinyl had to move out to the chimney breast. Moving the TV joins them into one 50.5 cm zone.
- **The turntable with the Series X beside it:** 45.2 + 20.5 cm won't go in 50.5. The Series X goes up on a shelf instead, above the lid's swing, where it gets more air than it had in v10 (15 cm each side, 21 above).
- **The vinyl cabinet on the breast as well as a table:** a table needs the breast's full width, and a cabinet in front would put the diners 44 cm further into the room.
- **A gateleg table in the recess:** it would block the door; the breast is the only clear wall.

Numbered view: the game drawer and the lower LP crate are pulled out 30 cm, with a view of each from above. Dashed lines show what's hidden: gold for data, audio and low-voltage leads, coral for mains. The sofa isn't drawn so the base stays in view; the room plan shows it.

OAK JOINERY	
1	Solid oak uprights, 4.5 × 30, worktop to ceiling (159 long)
2	The left upright is a corner post against the side wall
3	Flush wedged tenons carry the Series X shelf (134)
4	High shelf, top 196: books up to 30 cm
5	Record ledge above the TV: 4 LPs face-out
6	The oak run wraps the corner: 4 more LPs face-out
7	LED groove in each upright: a smart colour strip washes the wall
8	High shelf, right: a small framed sleeve
ON THE WORKTOP	
9	LG OLED C6 55in · centre 116.5 · now 25 cm nearer the side wall
10	Soundbar · eARC, optical and AUX inputs
11	Turntable (AT-LP120X size allowed) · lid clear to 130
12	Xbox Series X on its fan stand · 15 cm air each side
13	Phono preamp, hung under the shelf at the back
14	Solid oak worktop 186 × 46 × 3, top at 76
IN THE ALCOVE BASE	
15	Oak face frame; its bottom rail hides the 9 cm plinth
16	Xbox One + HDMI switch · its brick flat at the back
17	Xbox 360 + RetroTINK · its brick flat at the back
18	Game drawer: 2 face-out bins, spine-up rows behind
19	PS3 + controller basket · bay 2 top row
20	PS2 · bay 2, beside the sub
21	PS1 · under the PS2
22	Upper record tray: an LP crate, ~90 face-out, under the turntable
23	Lower LP crate: the same, shown pulled out
24	Slim drawer: pads and remotes, a charging lead
25	Two slim game drawers: ~22 games each, face-out
26	Hidden slim drawer in the plinth, finger pull
27	Sub · bay 2, under the TV, on an isolation riser

THE CHIMNEY BREAST (THE TABLE IS BY THE STAIRS)	
28	“Now playing” oak ledge: what's on and the next two
29	Two framed LPs above
THE SIDE WALL	
30	PC on the oak run, above the sofa's open end
31	Brush grommet: the PC's leads drop into the wall
32	Pad charging dock
33	PC cables chased into the wall: no spine, no trunking
34	Keyboard tray under the shelf, rolls out (v6.1)
POWER, NETWORK AND LIGHT	
35	Deco B: leads in the groove behind the right upright
36	Double socket: its right switch is the retro switch
37	10-gang extension, hidden behind the TV
38	LED strip round the ceiling edge (as now)
39	Smart spot bar on the existing rose, TV to window
40	Second smart spot bar, existing point by the door
41	Stairs UP to the first floor, beside the door (direction to confirm)

Built to your alcove: 186 wide, 38 deep, 235 to the ceiling

- **Base:** 3 cm filler, three 60 cm bays, 3 cm filler. Bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth (tops at 73, back panels off); bay 3 is an 18 mm ply carcass 60 × 40 × 73 from the floor, split by a 12 mm divider into the slim game column (two face-out game drawers and the pads drawer) and the LP column (two face-out LP crates) under the turntable. One oak face frame runs across all three; its bottom rail hides the plinth and a hidden drawer under bay 2 with a finger pull (no push-to-open at floor level).
- **Depth:** 40 cm frames, a 4 cm cable channel behind and the 2 cm face frame make 46 cm from the back wall, 8 cm proud of the chimney breast.
- **Worktop:** solid oak, 186 × 46 × 3, top at 76, with a brush-lined slot behind the TV and a grommet behind the deck.
- **Oak uprights:** 4.5 × 30 cm, 159 cm from the worktop to the ceiling. The left one stands against the side wall at 0–4.5, a corner post that frames the TV and carries the wrap ledge. The right one is at 131–135.5. That leaves 126.5 cm between them, 2.15 cm clear of the TV each side.

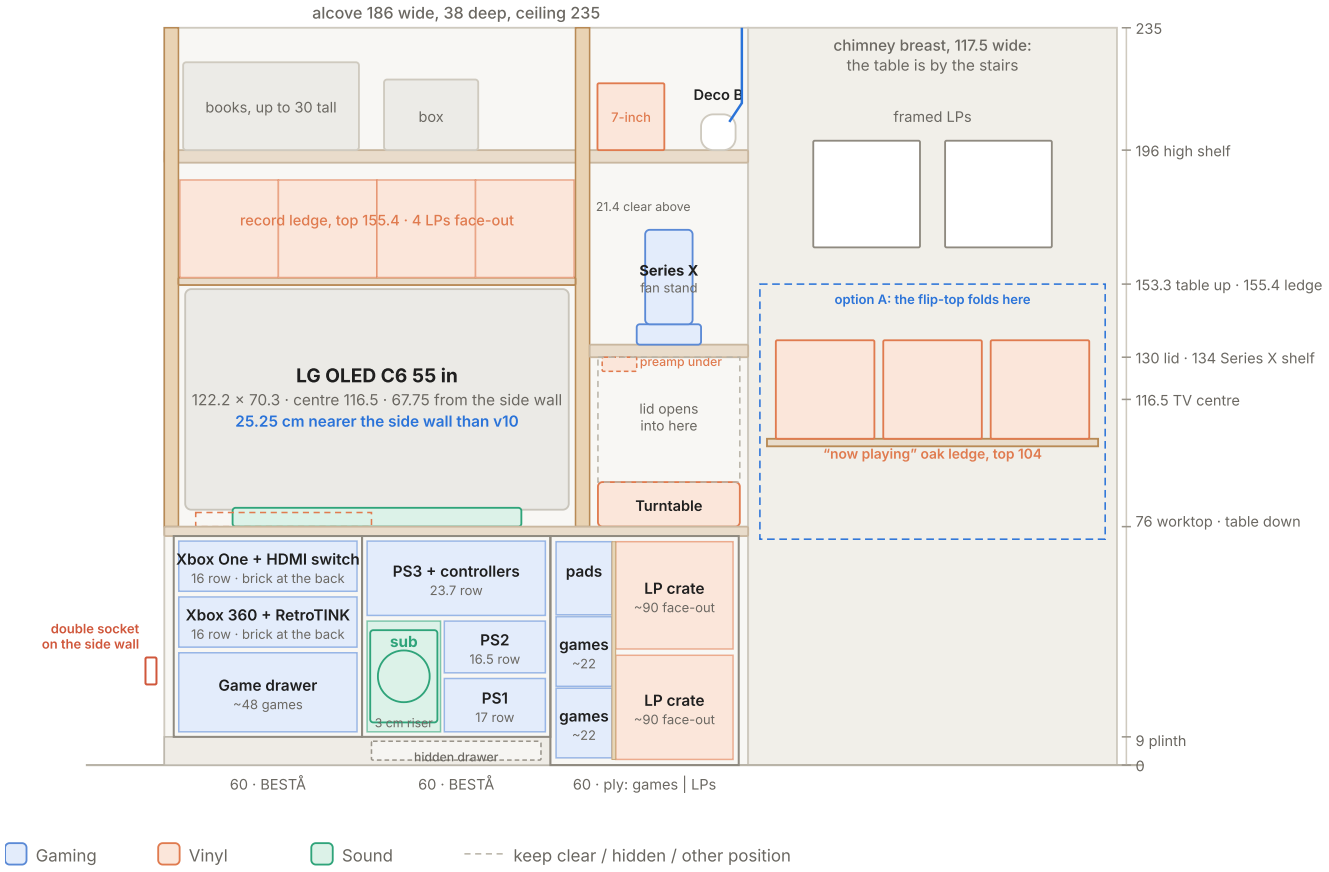
- **TV:** centre 67.75 cm from the side wall (it was 93), on LG's stand set 2 cm off the back wall: bottom 81.4, centre 116.5, top 151.7. The soundbar sits centred under it at the worktop's front edge.
- **The turntable zone:** 50.5 cm between the right upright and the breast. The deck sits on the worktop with 2.65 cm each side and its lid opens to about 121. Above it is an oak shelf at 134, through-tenoned into the upright, carrying the Series X on its fan stand; the phono preamp hangs underneath at the back.
- **Shelves above:** the record ledge at 155.4 between the uprights, four LPs face-out; the high shelf at 196, wall to wall in two lengths, with books, a box, a print, a small framed sleeve and Deco B.
- **Bay 2 with the sub (v13.5):** the PS3 keeps the top row. Below its shelf the bay splits: on the left the sub's column, 23.4 cm wide, then a 12 mm ply divider, then 32.2 cm for the PS2 over the PS1 on a part shelf. The sub stands on a 3 cm isolation riser with 1 cm each side and 2.9 above. Its front is flush with the face frame, so its driver faces the room and its rear port has 15.6 cm to the wall.
- **Power:** the side-wall socket in front of bay 1 becomes a double. Its two leads enter through a notch at the left end of the face frame's bottom rail and run back through the plinth to the channel.

The media wall, looked over again (v17)

- **The TV's remote:** the soundbar's top is 0.6 above the screen's bottom edge, right where LG puts the remote's receiver. Sit the bar 2 cm proud of the stand's foot so the beam from the sofa clears it; LG's Magic Remote works by radio once paired, so it doesn't mind either way (a check in the fit proof).
- **Heat:** every console bay is open front and back, the bricks lie flat at the back with nothing on top, and the Series X has 15 cm each side and 21 above. Nothing changed in v17.
- **One remote for most of it:** HDMI-CEC (SIMPLINK) lets the TV's remote work the soundbar and the modern consoles (step 13); the HDMI switch has its own remote for the retro side.

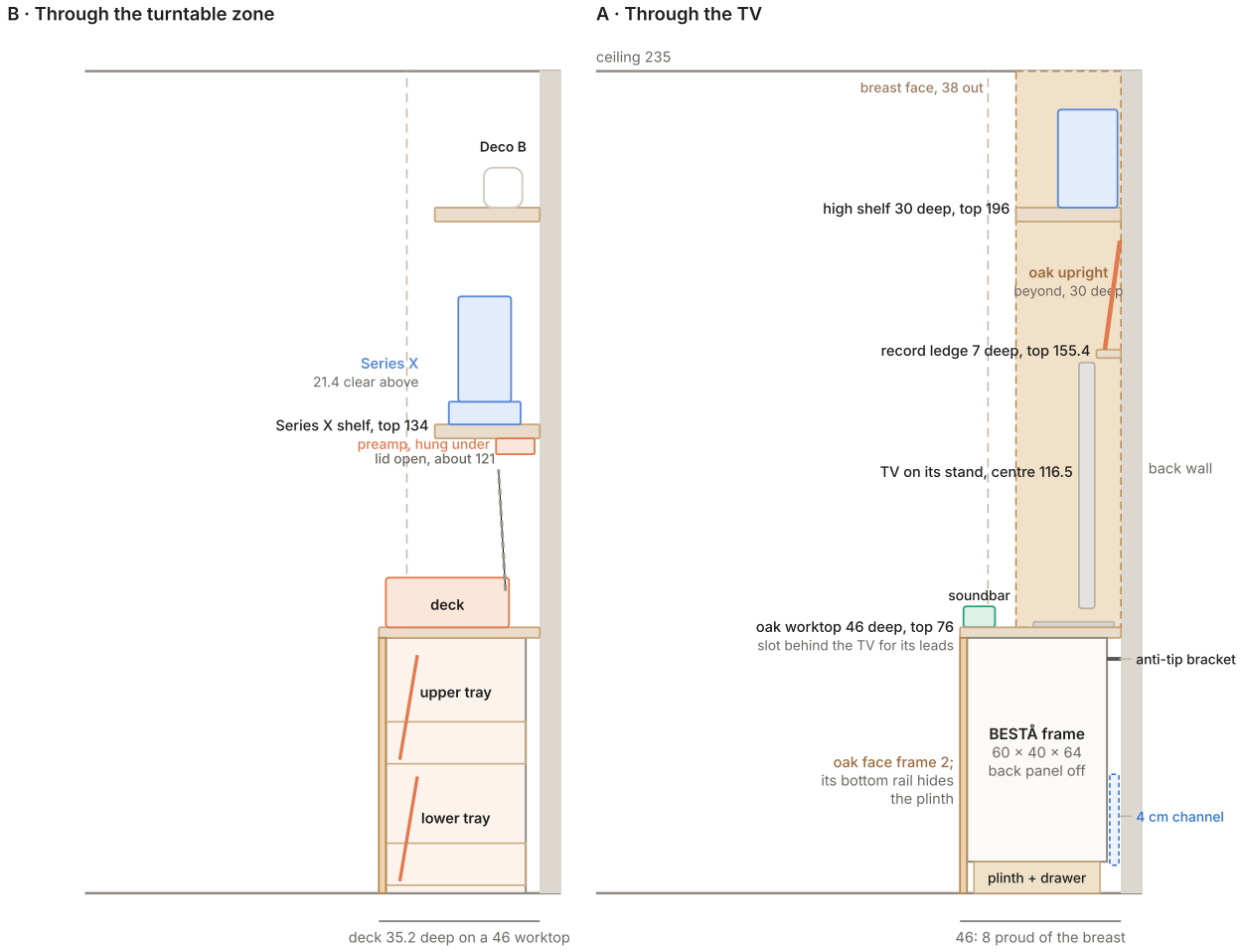
The media wall and the chimney breast from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below (bay 3: games beside the LP crates); the “now playing” ledg



How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



The sub in the media wall

You asked for the sub back in the unit, with the Xbox bricks moved behind. It fits, and it sits in the best spot for it: under the TV, beside the soundbar, so the bass comes from the same place as the picture. Here's how the base makes room, and what was checked.

- **The bricks go to the back.** The Xbox One's brick (17 × 7.5 × 4.8[11]) and the Xbox 360's (allowed 21 × 10 × 6[16]) lie flat at the back of their shelves, by the channel's power leads. Microsoft asks for a ventilated space, flat on the rubber pads, with nothing on top[26]; the bays are open front and back.
- **That frees bay 1's fronts** beside the two Xboxes, so the HDMI switch (beside the Xbox One, where its remote reaches it) and the RetroTINK (beside the Xbox 360) move up from bay 2.
- **Bay 2's lower half becomes two columns:** the sub on the left, under the TV, and the PS2 over the PS1 on the right. The sub is 21.4 × 30.4 × 29.4[3]; its column is 23.4 wide and 35.3 tall.
- **The port:** the sub's front sits flush with the face frame, so its rear port is 15.6 cm from the wall through the open back and the channel. That's inside the 15–30 cm advised for a rear port[25], at the low end, so the channel behind bay 2 stays clear: Switch 2 and the retro 6-way move behind bay 1.
- **Vibration, the one thing to test:** Microsoft advises keeping the Xbox 360 away from subwoofers[27], and the deck shares the worktop. The 360 is a whole bay away, the sub stands on an isolation riser without touching the divider or the frame, and the deck has isolation feet. Once it's built, play a disc game and a record at film volume. If anything skips, turn the sub down for that source; if it still does, the sub can go back under the dining wall.
- **The dining wall is clear underneath:** no cover to make, and nothing to move for eight at the table.

The fit proof: every item against its published size

Each zone below lists what goes in it, its size and where that size comes from (numbered sources, section 12), the space, what's left and a verdict. 59 of the 87 checks fit outright and none fail. 20 are tight: the deck's 2.65 cm sides and the preamp over the open lid; the sub's column, its port gap, the PS2 over the PS1 and the switch channel in bay 2; the LP crates, the game drawers and the pads drawer in bay 3, each snug by design; the flip-top's six at 53 cm each and its weight to fold up; the side chairs by the stairs; and, for option A, the coffee table and the door's swing. 8 need a check on site: the TV's plugs by the corner post, the RetroTINK's size when it arrives, whether the sub's vibration reaches the consoles and the deck (test it before fixing the shelves), that the soundbar sits clear of the TV's remote receiver (v17), what the stairs wall is built of, your shelf's height (the OBEGRÄNSAD needs it at 198 or lower to stand on it), and, for option A, the flip-top's end on the breast.

The TV between the uprights

126.5 between the oak uprights, the left one now a corner post on the side wall; the ledge's underside at 153

Item	Size (cm)	Space	Spare (cm)	Result
LG OLED C6 55 in, on its stand	122.2 × 70.3 × 4.5 panel; 75.7 tall on the stand[2]	126.5 wide	2.15 each side	**Fits**
Top of the screen	151.7 (bottom 81.4, centre 116.5)[2]	ledge at 153	1.3	**Fits**
LG stand base	47 × 23[2]	worktop 46 deep	set 2 cm off the wall: 11 cm in front of it for the soundbar's leads	**Fits**
Soundbar, Hisense AX5100G, centred under the TV	92 × 9 × 6 tall[3]	x 21.75–113.75, top 82	the sightline to the screen's bottom passes 83.54–84.17 over it	**Fits**
The TV's remote receiver, bottom centre of the screen's frame (v17 check)	screen bottom 81.4[2]	soundbar top 82, in front of the stand's foot	0.6 over the frame's edge: sit the bar on the worktop 2 cm proud of the stand's foot, not against it, so the remote's beam from the sofa clears it; if a button stutters, pair LG's Magic Remote (it talks to the TV by radio)	**Check**
The TV's HDMI plugs, with the screen 2.15 from the corner post	C6 sockets face sideways on the back panel[2]	behind the screen	plugs stay behind the panel; if a stiff lead pushes out, use 90° HDMI adapters	**Check**

The turntable zone, right of the TV

50.5 wide × 46 deep, worktop 76 to the Series X shelf's underside at 130

Item	Size (cm)	Space	Spare (cm)	Result
Turntable, AT-LP120X size (worst case)	45.2 × 35.2 × 14.2 with its lid[8]	50.5 wide	2.65 each side	**Fits, tight**
Its depth, with room behind for the lid's hinges	35.2, front 2 cm in from the worktop's edge[8]	46 deep, to the painted wall	8.8 behind it	**Fits**
Lid open	cover 44.1 × 34.6; standing up, its top at about 121[17]	shelf underside 130	about 9	**Fits**
Phono preamp, hung under the shelf at the back	about 11 × 11 × 4.5	125.5–130, at the back	4.4 over the open lid's edge; lift the lid gently	**Fits, tight**
Series X fan stand on the shelf	20.5 × 20.5 (assumed; model to confirm)	50.5 × 30 shelf	15 each side	**Fits**
Xbox Series X, standing	15.1 × 15.1 × 30.1 tall[4]	top at 170.6	21.4 clear above (Microsoft's minimum is 4); 15 each side, inside the 10–15 it recommends	**Fits**
Deco B on the high shelf, right	11 × 11 × 11.4[7]	50.5 wide, 39 to the ceiling	beside the framed 7-inch, 27 clear above	**Fits**

Shelves above the TV

ledge top 155.4, 7 deep; high shelf top 196, 30 deep; ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Four LPs on the record ledge	4 × 31.4 = 125.6 wide[13]	126.5 long	0.9: snug, as drawn	**Fits**
Ledge LPs leaning back 8°	tops at 186.5[13]	high shelf underside 192	5.5 to lift one out over the 1.5 lip	**Fits**
Books on the high shelf	up to 30 tall	39 to the ceiling	9 over the tallest	**Fits**
Oak uprights	159 long (worktop to ceiling)[1]	235 ceiling	buy 165 and scribe to the ceiling	**Fits**

Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)

56.8 wide inside, 40 deep; the bricks lie at the back of their rows

Item	Size (cm)	Space	Spare (cm)	Result
Xbox One, original, + HDMI switch	33.3 × 27.4 × 7.9; switch 12.7 × 5.9 × 2.5[12,10]	16 tall row	8.1 above; the switch in the 23.5 beside it, at the front for its remote	**Fits**
Xbox 360, original, + RetroTINK-6X CE	30.9 × 25.8 × 8.3; RetroTINK allowed 23 × 23 × 8 (no size published)[12,9]	16 tall row	7.7 above; 25.9 beside it for the RetroTINK: measure it when it arrives	**Check**
Both bricks, flat at the back of their rows	Xbox One 17 × 7.5 × 4.8; Xbox 360 allowed 21 × 10 × 6[11,16,26]	behind the switch and the RetroTINK, by the channel's power leads	23 + 10 = 33 of the 40 depth for the RetroTINK and the 360's brick; on their rubber pads, nothing on top, open to the channel; 2 cm or more from the consoles' rear vents	**Fits**
Game drawer	box 54.3 × 35 on 350 mm runners; cases 19 × 13.5 × 1.4[15,13]	25.2 tall row, 40 deep	about 3 above the leaning cases; 5 behind the box	**Fits**
Depth with the leads	console 27.4 + HDMI plug about 5[12]	40 deep, back open	about 7.6, then the 4 cm channel	**Fits**

Bay 2, PlayStation and the sub (BESTÅ frame on the plinth)

56.8 wide inside, 40 deep; under the PS3's shelf, the sub's column 23.4 wide, a 12 mm ply divider, then 32.2 for the PS2 over the PS1

Item	Size (cm)	Space	Spare (cm)	Result
PS3, original, + controller basket	32.5 × 27.4 × 9.8 [12]	23.7 tall row, full width	13.9 above; 24.3 beside for the basket	**Fits**
The sub in its column	21.4 × 30.4 × 29.4 tall, on a 3 isolation riser [3]	23.4 wide × 35.3 tall	1 each side, 2.9 above; it mustn't touch the divider or the frame	**Fits, tight**
The sub's driver and port	front-firing driver, rear port; its front flush with the face frame [3,25]	15.6 from the port to the wall, through the open back and the channel	inside the 15–30 advised for a rear port, at its low end: keep the channel behind bay 2 clear (Switch 2 and the retro 6-way go behind bay 1)	**Fits, tight**
PS2, original, over the PS1	30.2 × 18.3 × 7.8 · 27 × 18.8 × 6 [12]	32.2 wide; 16.5 and 17 rows on a part shelf	PS2 1 each side, 8.7 above; PS1 2.6 each side, 11 above; both open front and back	**Fits, tight**
Vibration: the consoles and the deck share the base with the sub	Microsoft: keep the Xbox 360 away from high-vibrating sources such as subwoofers [27]	the sub on its riser, clear of the divider and the frame; the deck on isolation feet	the 360 is a bay away, the PS2 and PS1 a divider away: play a disc game and a record at film volume; if anything skips, turn the sub down for that source, or put it back under the dining wall	**Check**
Hidden slim drawer in the plinth	box 52 × 34 × 5, 300 mm full-extension runners, a finger-pull groove (no push-to-open at floor level) [15]	9 cm plinth, 120 wide	about 2 above and below the box; felt its runners, it's under the sub	**Fits**
Switch 2 (TL-SG108) and the retro 6-way in the channel	15.8 × 10.1 × 2.5; 6-way about 34 × 6 × 4 [6]	4 cm channel, behind bay 1	on their edges, clear of the sub's port	**Fits, tight**

Bay 3: the LP crates and the slim game column (v16)

56.4 wide × 40 deep × 69.4 tall inside the ply carcass, split by a 12 mm divider

Item	Size (cm)	Space	Spare (cm)	Result
Across the bay	slim drawer 15.26 + divider 1.2 + LP crate 34.86, each drawer with 1.27 runners a side [15]	56.4 inside	exact: 15.26 + 2.54 + 1.2 + 34.86 + 2.54 = 56.4; nothing left over	**Fits**
LPs in the crate	31.4 sleeves face-out, leaning back 10° [13,36]	33.06 trough	0.83 each side: snug like a shop crate; outer sleeves up to 32.4 still fit	**Fits, tight**
Crate heights	lower sleeves to 34.4, upper to 69.1 [13]	top panel underside 71.2	2.6 and 2.1 to lift a sleeve	**Fits**
Games in the slim drawers, face-out	DVD-size cases 13.5 wide, 19 tall, leaning back 10° [13]	14.06 inside (6 mm ply sides)	0.56 spare: they flip like the records; about 22 a drawer in 40 deep	**Fits, tight**
The game drawers' height	cases stand 18.71 on a 1.2 base [13]	22.8 from one drawer to the next	2.89 clear over the cases	**Fits**
Pads in the top slim drawer	4 pads on edge, two side by side: 2 × 6.6 = 13.2 across, 16 long [35]	14.06 wide, 22.1 tall	0.86 across; 10.6 tall in 22.1; the remotes in front	**Fits, tight**

The side wall: one oak run from the corner to the PC

one board line at 155.4, 4 thick, 220 long: 7 deep along the LPs, sweeping out to 28 under the PC (v15)

Item	Size (cm)	Space	Spare (cm)	Result
The sweep from the ledge to the PC shelf	front edge 7 → 28 over 14 (z -118 to -132)	the last wrap LP ends at z -123.3; the PC starts at z -134	the LP stands on the start of the curve; 2 between the full depth and the case	**Fits**
Four more LPs on the wrap ledge	4 × 31.4 + gaps = 129.8 [13]	134 long	2.7	**Fits**
The PC on its shelf	23 × 45.3 × 46.6 tall (an average gaming case, e.g. Corsair 4000D) [29]	28 × 82	2.5 off the wall, the brush grommet and pad dock beyond it; height 46.6 under the 79.6 to the ceiling	**Fits**
Air above the PC	top at 202	235 ceiling	33 for the top exhaust	**Fits**
Shelf over the end of the sofa	underside 151.4 [24]	sofa back 80	71.4; the shelf covers only the 28 cm nearest the wall	**Fits**
Keyboard tray under the PC shelf (v6.1)	7 cm housing, tray rolls 25 out on 300 mm runners [15]	sofa back 80	64.4 over the sofa's back; pulled out, its front edge is 53 from the wall, still over the return's seat	**Fits**
Out of the TV's sightlines	PC about 54° left of the line to the screen from the sofa [1]	—	v10's PC sat at the screen's left edge	**Fits**

The flip-top table under your shelf, by the stairs (B, the default)

110 × 160 open, running from the stairs wall into the room, its end at x 296.7

Item	Size (cm)	Space	Spare (cm)	Result
Open: 110 × 160, still on the wall	the top 110 × 80 on its brackets, the half 110 × 80 on the gate leg [18,30]	nothing comes off the wall; no leaf to store	the table runs 170.3 out from the wall	**Fits**
Three flip-top hinges at the joint	1.27 × 7, set into both edges [30]	a 2.7 edge	0.72 of oak either side of each hinge; they open to 180°, so the half lies flat on the top to fold	**Fits**
The gate leg folds inside the half	legs 73.3 long, hinged 3 in from the far edge [20]	74.75 of the half for them to lie along	1.45 spare	**Fits**
Folded on the wall	the top, the half and its gate: 13.3 off the wall, top edge 153.3 [19]	a 7.3 hinge rail makes room for the half and gate behind the top	about 7 deeper than v14's; the room still sees the plain oak underside	**Fits**
Folding it up	top 17 kg + half 17 kg + gate 2 kg [19,18]	lift-off hinges for a 120 kg door; brackets 500 kg a pair	about 37 kg to swing up: two people, one at each corner; turn buttons hold it	**Fits, tight**
Knees by the brackets	each arm 15 in from the side edge, at 70.9 [18,22]	knees at about 60 over a 48 seat	the first chair each side has the arm above its knees, not in them	**Fits**
Four: two down each side	80 of edge each [23]	60 each	20 spare each	**Fits**
Six: three down each side	53.33 of edge each [23]	60 each, 55 at a squeeze	53.33 each: elbows in	**Fits, tight**
The stair side's chairs	backs at z -84 when pulled out [22,28]	the stair box's face at -41	43 behind them: tuck them in after a meal so the stair's foot stays clear	**Fits, tight**
The window side's chairs	backs at z -276 [22]	the window wall at -309	33 behind them; your red lamp moves from the window corner to the sofa's end or by the breast while it's open	**Fits, tight**
A seat at its end	one chair at x 255.7–304.7 [22,24]	the sofa's arm ends at x 268, front z -201	one chair, centred on the table, meets the sofa's front (0.5 over, with the sofa 13 off the wall since v17): tuck it 2 cm further under the table: seven with three a side (a second end chair would hit the arm)	**Fits, tight**
Folded, under the shelf	top edge 153.3, 13.3 off the wall [28]	shelf's underside about 191	37.7: the shelf and what's on it stay	**Fits**
Light over the table	the existing pendant point, for bar 2 [28]	point about (412, -143)	over the open table's stair side: the bar runs along it	**Fits**
Your shelf and the pipe boxing (v16)	the shelf 200 long [28,1]	the boxing's face at z -286, 22 out	drawn 2 short of it, -43 past the stair box (your photo: about 41)	**Fits**
The OBEGRÄNSAD panel on the shelf (v16)	52 × 36 × 6, standing landscape at the stair end [34]	shelf top 196 to the 235 ceiling	3 over it: this needs the shelf at 198 or lower (measure it); higher, hang the panel on the wall beside the shelf instead	**Check**
The window side's chairs and the pipe boxing	backs at z -276 [22,1]	the boxing's face at z -286 (x 445 to the wall)	10 clear of the boxing	**Fits**
Fixing the brackets	500 kg a pair, into the wall [18]	the west wall's build isn't known yet	solid (plugs and screws) or studwork (fix into studs): knock and look before choosing this site	**Check**

The chimney breast, with the table by the stairs (v16)

breast 117.5 wide; the “now playing” ledge 105.5 × 7, top 104

Item	Size (cm)	Space	Spare (cm)	Result
Three LPs on the ledge	3 × 31.4 = 94.2 [13]	105.5 long	2.83 between and at the ends	**Fits**
Leaning back 8°, under the framed LPs	tops at 135.09 [13]	frames from 165	29.91 between	**Fits**
Beside the turntable	ledge 104 up, 7 deep [8]	the deck's top at about 90.2, an arm's reach to the left	sleeve off the ledge, record on, sleeve back: the base stands 8 proud of the breast, the ledge 7	**Fits**
If you choose option A instead	the flip-top folds on the breast [19]	the ledge's place	the ledge comes off its two screws; the frames stay	**Fits**

Option A: the flip-top table on the chimney breast

110 × 160 open, running from the breast towards the window, 170.3 out

Item	Size (cm)	Space	Spare (cm)	Result
Open: 110 × 160, still on the wall	the top 110 × 80 on its brackets, the half 110 × 80 on the gate leg [18,30]	nothing comes off the wall; no leaf to store	the table runs 170.3 out from the wall	**Fits**
Three flip-top hinges at the joint	1.27 × 7, set into both edges [30]	a 2.7 edge	0.72 of oak either side of each hinge; they open to 180°, so the half lies flat on the top to fold	**Fits**
The gate leg folds inside the half	legs 73.3 long, hinged 3 in from the far edge [20]	74.75 of the half for them to lie along	1.45 spare	**Fits**
Folded on the wall	the top, the half and its gate: 13.3 off the wall, top edge 153.3 [19]	a 7.3 hinge rail makes room for the half and gate behind the top	about 7 deeper than v14's; the room still sees the plain oak underside	**Fits**
Folding it up	top 17 kg + half 17 kg + gate 2 kg [19,18]	lift-off hinges for a 120 kg door; brackets 500 kg a pair	about 37 kg to swing up: two people, one at each corner; turn buttons hold it	**Fits, tight**
Knees by the brackets	each arm 15 in from the side edge, at 70.9 [18,22]	knees at about 60 over a 48 seat	the first chair each side has the arm above its knees, not in them	**Fits**
Four: two down each side	80 of edge each [23]	60 each	20 spare each	**Fits**
Six: three down each side	53.33 of edge each [23]	60 each, 55 at a squeeze	53.33 each: elbows in	**Fits, tight**
Its end and the sofa	the table ends 170.3 out [24]	the sofa's front at 201	30.7 between them: no chair at that end (it would need 41), and the way round is down the sides	**Check**
The coffee table	east-side chairs back to x 148.75 [22]	coffee table x 103–193	for six, slide the coffee table towards the TV before you sit down	**Fits, tight**
The door's swing	door 83 wide, opens in, hinged by the breast [1]	its edge sweeps 83 round the hinge	the west side's first chair is 60.03 from the hinge: in its swing, so keep the door shut while you eat (or rehang it to open into the hall)	**Fits, tight**

The room

467 × 347, ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Eyes to screen from the sofa	the L-sofa's long side against the window wall, 10 off the radiator [1,24]	335 window wall to screen	about 2.65 m; a 55-inch fills about 26°	**Fits**
From the end of the sofa's return	sitting up at its end, 214 from the screen [24]	straight ahead of the TV	the TV's move puts it square on to this seat	**Fits**
TV centre against seated eyes	116.5 [2]	eyes at 105–112	about 1–2.5° above the eye line	**Fits**
Both spotlight bars under the ceiling	22.4 drop [14]	235 ceiling	lowest shade at 212.6	**Fits**
Both bars turned to run TV to window (v15)	84 long along the room [14,28]	bar 2 at 55 from the west wall	bar 1 spans z -62.5 to -146.5; bar 2 now runs along the wall, so its end no longer reaches past the west wall (v13.7's worry)	**Fits**
Smart colour bulbs in the bars (v15)	8 × E14: WiZ P45 4.9 W, 4.5 × 10.1; or Hue lustre 5.1 W, 4.5 × 7.7 [14,38]	the bar's 6 W a lamp; 15 cm shades	1.1 W under; the WiZ's 10.1 sits inside a 15 shade with its holder; if it shows below the rim, the Hue's 7.7 doesn't	**Fits**

Your plant stand and your record shelf (v18)

the window wall's west end, 107 between the curtains' stack and the pipe boxing; the chimney breast, 117.5 wide

Item	Size (cm)	Space	Spare (cm)	Result
A: the plant stand in the window corner (recommended)	90 × 30 × 120 [32]	x 338 to 445	x 345–435: 7 from the curtains, 10 from the boxing; daylight from the window beside it, its grow bars for the rest	**Fits, tight**
A: the 2 × 2 record shelf on the breast	77 × 39 × 77, four trays [33]	breast 117.5 wide, the “now playing” ledge's underside at 101.6	20.25 each side, 24.6 under the ledge; a step from the turntable	**Fits**
A: records in the 2 × 2	four trays, face-out, about 40 LPs each [33,13]	33 × 33 cubes	about 160 more LPs, flipped like the crates	**Fits**
B: the record shelf in the window corner, as today	77 wide [33]	107 between the curtains and the boxing	30 to spare	**Fits**
B: the plant stand on the breast	90 wide, 120 tall [32]	breast 117.5	13.75 each side; the “now playing” ledge comes off (the plants stand above its height), the frames stay above	**Fits, tight**
Power for the grow bars	a USB or mains lead from the stand [32]	the nearest socket	A: your photo shows an extension lead in that corner today; put the grow bars on a smart plug with a schedule (about 12 hours a day). B: the double socket by bay 1 is a few steps away	**Check**
Your red lamp	28 base [1]	the window corner is taken in A and B	it moves to the sofa's west end (in the 3D room, drag it anywhere)	**Fits**

Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

Console	Version	W × D × H (cm)	Power
PS1	Original (largest) · PS one	27.0 × 18.8 × 6.0 · 19.3 × 14.4 × 3.8	Built in · external adapter
PS2	Original (largest) · Slim	30.2 × 18.3 × 7.8 · 23.1 × 15.2 × 2.8	Built in · external brick
PS3	Original (largest) · Slim · Super Slim	32.5 × 27.4 × 9.8 · 29.0 × 29.0 × 6.5 · 29.0 × 23.0 × 6.0	Built in
Xbox 360	Original (largest) · S · E	30.9 × 25.8 × 8.3 · about 27.0 × 26.4 × 7.5 · 25.4 × 26.4 × 6.7	External brick
Xbox One	Original (largest) · S · X	33.3 × 27.4 × 7.9 · 29.5 × 22.9 × 6.4 · 30.0 × 24.0 × 6.0	External brick · built in
Xbox Series X	Standing	15.1 × 15.1 footprint, 30.1 tall	Built in

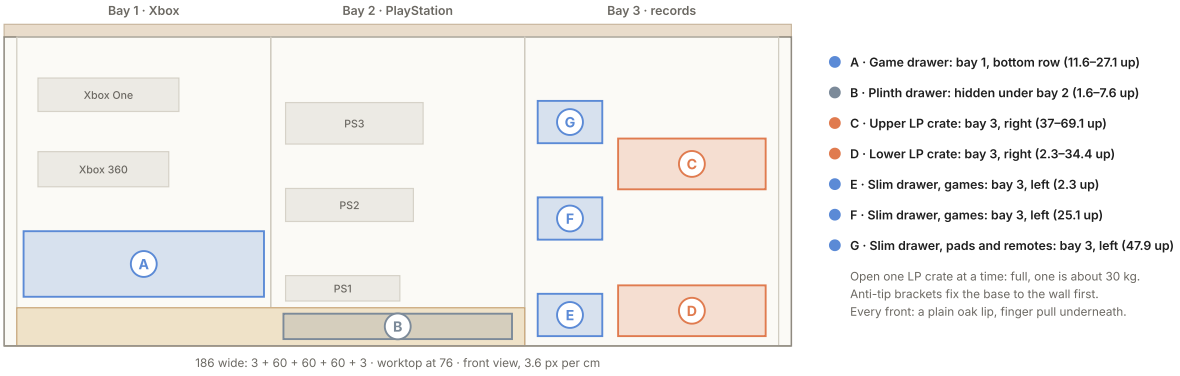
The drawer layouts

Seven drawers in the alcove base, all drawn to scale from above: **A** the game drawer (bay 1), **B** the hidden plinth drawer under bay 2, **C** and **D** the two face-out LP crates and **E** to **G** the slim drawers beside them (bay 3). Each one shows its compartments, what goes in them and how many, and its runners. Every front is a plain oak lip with a finger pull cut underneath.

The drawer layouts: where each drawer is and what goes in it

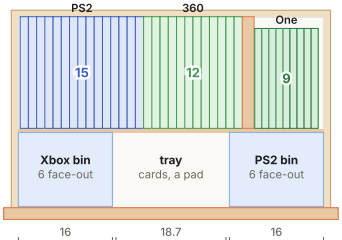
Seven drawers in the alcove base, all to scale: A the game drawer, B the hidden plinth drawer, C and D the LP crates, E to G the slim drawers beside them

Where each drawer is: the alcove base from the front



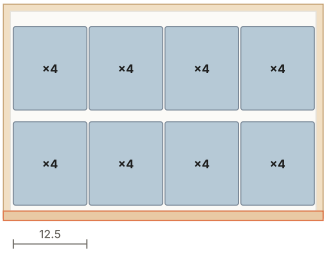
186 wide: 3 + 60 + 60 + 60 + 3 · worktop at 76 · front view, 3.6 px per cm

A · Game drawer
box 54.3 × 35 · 350 mm soft-close runners · rolls out 30



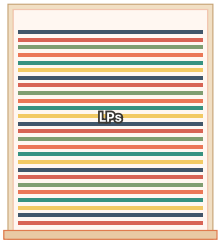
back: 36 spine-up, titles on top (19 cm cases on edge)
front: 12 face-out · about 48 games in all

B · Hidden plinth drawer
box 52 × 34 × 5 inside · 300 mm runners · finger pull



32 PS1 games flat: 8 stacks of 4 jewel cases
or remotes, spare pads, cables and manuals

C and D · LP crates
box 34.9 × 40 · 400 mm runners



trough 33.1: snug, no dead space
face-out, leaning back 10°
~90 LPs each · two crates

E, F, G · Slim drawers
15.3 × 40 · 400 mm

E, F: ~22 games
each, face-out
G: 4 pads, remotes,
a charging lead

Plans drawn from above at 5 px per cm, front (the room) at the bottom; the elevation at 3.6 px per cm. Sizes: BESTÅ frame 60 × 40 × 64 (IKEA);
game cases 19 × 13.5 × 1.4 and 17 × 13.5 × 1.2; jewel case 14.2 × 12.5 × 1; LP sleeve 31.4; pads 16 × 10.6 × 6.6; runners Emuca 400 mm, Accuride 350 mm (see the fit proof).

Every drawer out, in place: the game drawer, the upper LP crate and the middle slim game drawer 30 cm out, the plinth drawer 25 cm out. In use, open one LP crate at a time (a full one is about 30 kg).

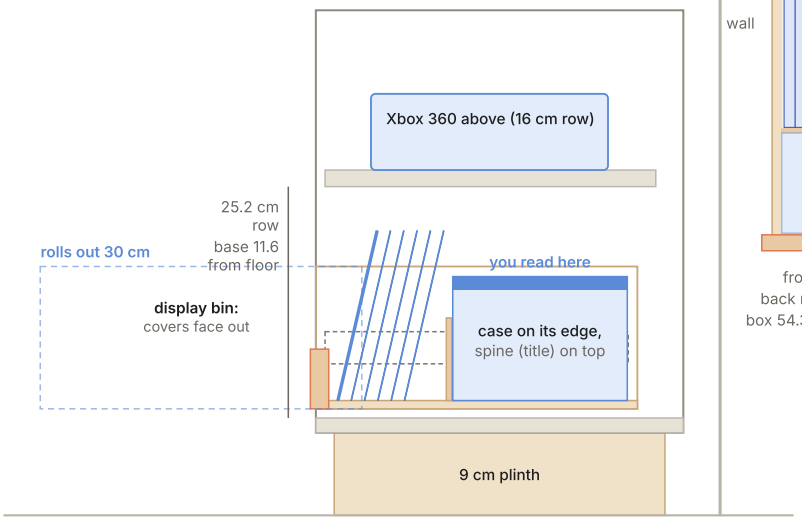
The game drawer (bay 1, bottom row)

Unchanged. Two display bins at the front show covers face-out even with the drawer shut (Xbox left, PS2 right, a small tray between them); behind them the rest of the games stand on edge with the spine on top, so every title faces up when you roll it out.

The game drawer: covers on show at the front, every title readable from above at the back

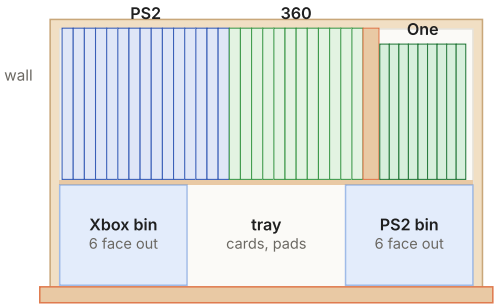
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out



front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

- **Box:** about 54.3 × 35 cm outside on 350 mm full-extension soft-close runners screwed to the BESTÅ sides; sides 15 cm; 9 mm base lined in felt. About 48 games.
- **Runners:** Accuride DZ3832EC (43 kg) or Screwfix Essentials 964VN (40 kg); not IKEA's BESTÅ runners, which aren't full extension.
- **The bricks:** both Xbox bricks lie flat at the back of their shelves, on their rubber pads, beside the channel's power leads, with nothing on top. Microsoft only asks that the space is ventilated[26], and these bays are open at the front and the back.

4 · The records and the turntable

Your plant stand and your 2 × 2 record shelf (v18)

You've bought a plant stand (two columns, four slatted tiers, warm grow lights under the top two) and you're keeping your 2 × 2 record shelf with its four trays. There are two good places for them, the window's west corner and the chimney breast (free now the table is by the stairs), so there are two layouts. In the 3D room the “Your plant stand and record shelf” buttons switch between them, and in Edit room you can drag either anywhere.

- **A, recommended: plants in the window corner, records on the breast.** The stand stands against the window wall between the curtains' stack and the pipe boxing (107 cm, with 7 and 10 to spare), so its plants get daylight from the window beside them and the grow lights for the rest. The record shelf stands on the breast, centred, under the “now playing” ledge (24.6 cm clear): pick a record from a tray, play it on the deck an arm's reach away, stand its sleeve on the ledge. About 160 more LPs, face-out like the crates.
- **B: records in the window corner, as today, plants on the breast.** The stand's grow lights face the sofa and the plants become the breast's display; the “now playing” ledge comes off (the plants stand above its height) and the two frames stay above.
- **Either way:** your red lamp moves from the window corner to the sofa's west end; the grow lights go on a smart plug with a schedule (about 12 hours a day); drip trays or a waterproof mat under the pots protect the slats and the carpet; one anti-tip strap holds the record shelf to the wall (about 35 kg of records). Both are drawn at assumed sizes (the stand 90 × 30 × 120, the shelf 77 × 39 × 77 like IKEA's KALLAX 2 × 2) until you measure them.

Layout A, from the 3D room: your plant stand in the window's west corner, its grow lights glowing under the top two tiers.

Layout A: your 2 × 2 record shelf on the chimney breast, under the “now playing” ledge, beside the turntable.

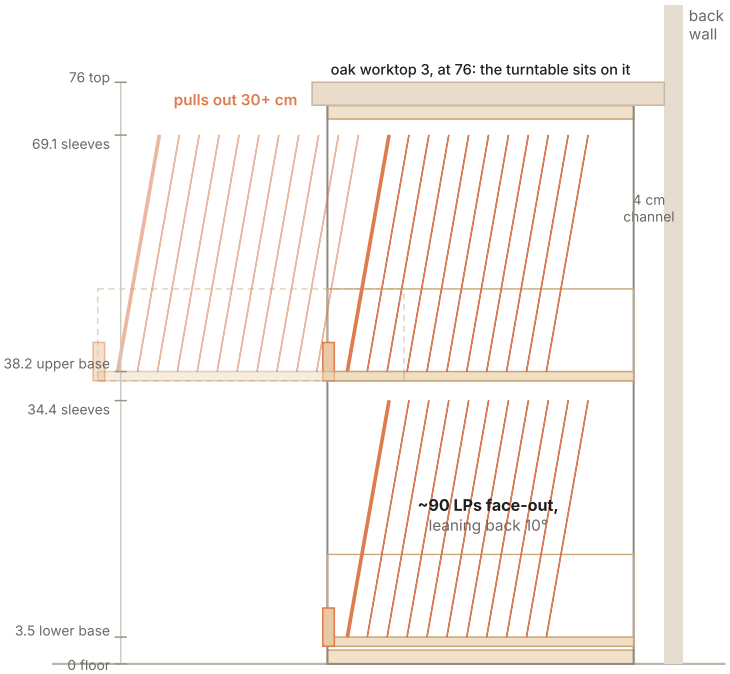
Layout B: the plant stand on the breast instead; the record shelf goes back to the window corner.

The vinyl is in one place, in the alcove: the deck on the worktop, the records in the two LP crates right underneath it, the preamp tucked under the shelf above, and (v16) the “now playing” ledge on the breast an arm's reach away. Pull a crate, pick a record, play it, and stand its sleeve on the ledge.

Bay 3: two LP crates under the turntable, a slim game column beside them

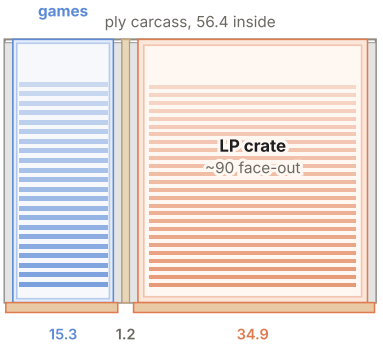
Section left through the LP crates, plan right. Pull a crate and flip through the covers like a record shop browser; the slim drawers flip the same way for games

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · crates 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Bay 3 from above (v16)



LP crate 34.9 × 40, a 33.1 trough: LPs snug, no dead space
slim drawers 15.3 × 40, 14.1 inside: DVD-size cases face-out
12 mm ply divider · 12.7 mm runners each side
an LP crate is about 30 kg full: open one at a time

How bay 3 works (v16)

- **Why it changed:** with LPs only, v15's trays had about 10 cm of empty width each side. Now a 12 mm divider splits the bay: the LP crates take only the width LPs need, and the rest becomes a slim column that does a job.
- **The LP crates:** a 33.06 cm trough, the snug width record shops use[36]: the sleeves stand with a little play and you flip them by their top edges. They lean back 10°; the last one leans on a raked oak rest. Outer sleeves up to 32.4 still fit.
- **Capacity:** about 90 LPs per crate; two crates, about 180 LPs, the same as before.
- **The slim column:** three drawers 15.26 wide, 14.06 inside. The bottom two hold DVD-size game cases face-out (13.5 wide), about 22 each, flipped like the records. The top one holds four pads on edge[35], the TV, soundbar and streaming remotes, and a USB-C lead fed from the 10-gang to charge a pad in the drawer.
- **Runners:** 400 mm full extension everywhere; 68–70 kg a pair under the crates. A full crate weighs about 30 kg: open one at a time, and fit the anti-tip brackets before loading.

Dimension	Figure	How it adds up
Bay 3 carcass	60 × 40 × 73 outside, 56.4 × 69.4 inside	18 mm birch ply; back open to the channel; a 12 mm divider 17.8 in from the left
Across the bay	15.26 + 2.54 + 1.2 + 34.86 + 2.54 = 56.4	Slim drawer, its runners, the divider, the LP crate, its runners: nothing left over
LP crate	34.86 × 40 outside, 33.06 × 38.2 inside	9 mm sides and ends; 0.8 either side of a 31.4 sleeve
Slim drawers	15.26 × 40 outside, 14.06 inside	6 mm sides; bases at 2.3, 25.1 and 47.9
Lower crate · upper crate	Undersides 2.3 · 37.0; sleeves to 34.4 · 69.1	2.6 and 2.1 cm finger room; top panel at 71.2

The turntable zone

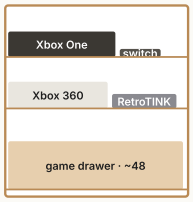
- **The deck:** on isolation feet, 2 cm back from the worktop's front edge, with 8.8 cm behind it for the lid's hinges. The zone takes a deck up to the size of an AT-LP120X (45.2 × 35.2 cm); most decks are smaller.
- **The lid:** AT's cover is 44.1 × 34.6 cm[17], so standing open its top edge is at about 121; the shelf's underside is at 130.
- **Bass:** the sub is in bay 2, in the same base the deck sits on. It stands on an isolation riser, clear of the divider and the frame, and the deck stands on isolation feet. If a record skips or hums at film volume, turn the sub down for vinyl.
- **Signal path:** turntable set to PHONO → the preamp under the shelf → a 2 m RCA-to-3.5 mm lead down the groove behind the right upright and along the back of the worktop to the soundbar's AUX. If the deck has no PHONO/LINE switch, plug it straight into the AUX.
- **Soundbar:** Music mode, bass up, surround, night and dialogue modes off. Always the AUX cable, never Bluetooth.

Storage options, to scale

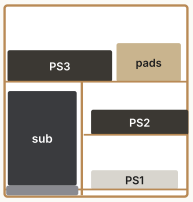
Storage options: what fits each space, to scale

Front views and views from above of every storage space in the wall, with counts, and where the overflow would go

Bays 1 and 2: inside each BESTÅ frame
56.8 wide × 40 deep inside; rows as planned, front view



Bay 1 · Xbox



Bay 2 · PlayStation + sub

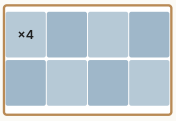
Swap any 16 cm row for two KUGGIS boxes, 26 × 35 × 15, side by side (52 of 56.8 wide).

Bay 3 from above: LP crate and slim drawer
56.4 inside: a 15.3 slim drawer, the divider, a 34.9 LP crate

games | divider | LPs: no dead space

≈ 90 LPs a crate, two crates
= 180 LPs, all face-out
beside them 44 games face-out
and a drawer for pads and remotes
trough 33.1: thick outer sleeves fit

Hidden plinth drawer, from above
box 52 × 34 × 5 inside, under bay 2; finger pull, not push-to-open



32 PS1 games flat
jewel cases 14.2 × 12.5 × 1, 8 stacks of 4
or: remotes, spare pads, cables, manuals

Series X shelf, front view
50.5 wide, 134 to the high shelf at 192: 58 clear

15 air each side, 21.4 above
Microsoft: at least 4 above, 10–15 recommended all round
Under it: the deck, lid opens to 121

High shelf, front view: as planned, or as LP overflow
186 long, 30 deep, 39 clear to the ceiling; the right-hand 50.5 keeps Deco B and the framed 7-inch

about 1 m of books, a lidded box, a print

As planned

≈ 380 LPs spine-out at about 3.3 mm each; a 31.4 sleeve overhangs the 30 cm shelf by 1.4

As LP overflow

PC shelf, from above
82 along the wall × 28 out, at 155.4; keyboard tray in a 7 cm housing under it

side wall along the top · alcove to the right

Keyboard tray: 300 mm runners, rolls 25 cm out; the housing stops 12 short of the sofa end so the cables drop past it. Full-size wireless keyboard and mouse.

If the collection outgrows the wall
two overflow options, both face-out or spine-out to match

1 · LPs spine-out on the high shelf: about 380 more
No building; books move to the second option's top.

2 · A second record unit, west of the sofa, under the window
Another LP column like bay 3 (40 × 46 × 73, two crates): 180 LPs.
At x 300–360 it leaves 123 cm to the table set for six (100 to walk).

front view, 60 × 73 (shown at half height)

All to one scale, 2.2 px per cm. Sizes: KUGGIS box 26 × 35 × 15 (IKEA); jewel case 142 × 125 × 10 mm (Wikipedia: Optical disc packaging); consoles, sleeves, Series X and the trays from the fit proof (section 3 of the plan).

Where every record lives

Where	What	How many
Bay 3, the two LP crates	LPs, face-out	About 180 LPs
The breast (v16)	The “now playing” ledge	3 LPs, the one that's on and the next two
Above the TV	Oak record ledge, the "now playing" wall	4 LPs
Side wall	The oak run round the corner	4 LPs

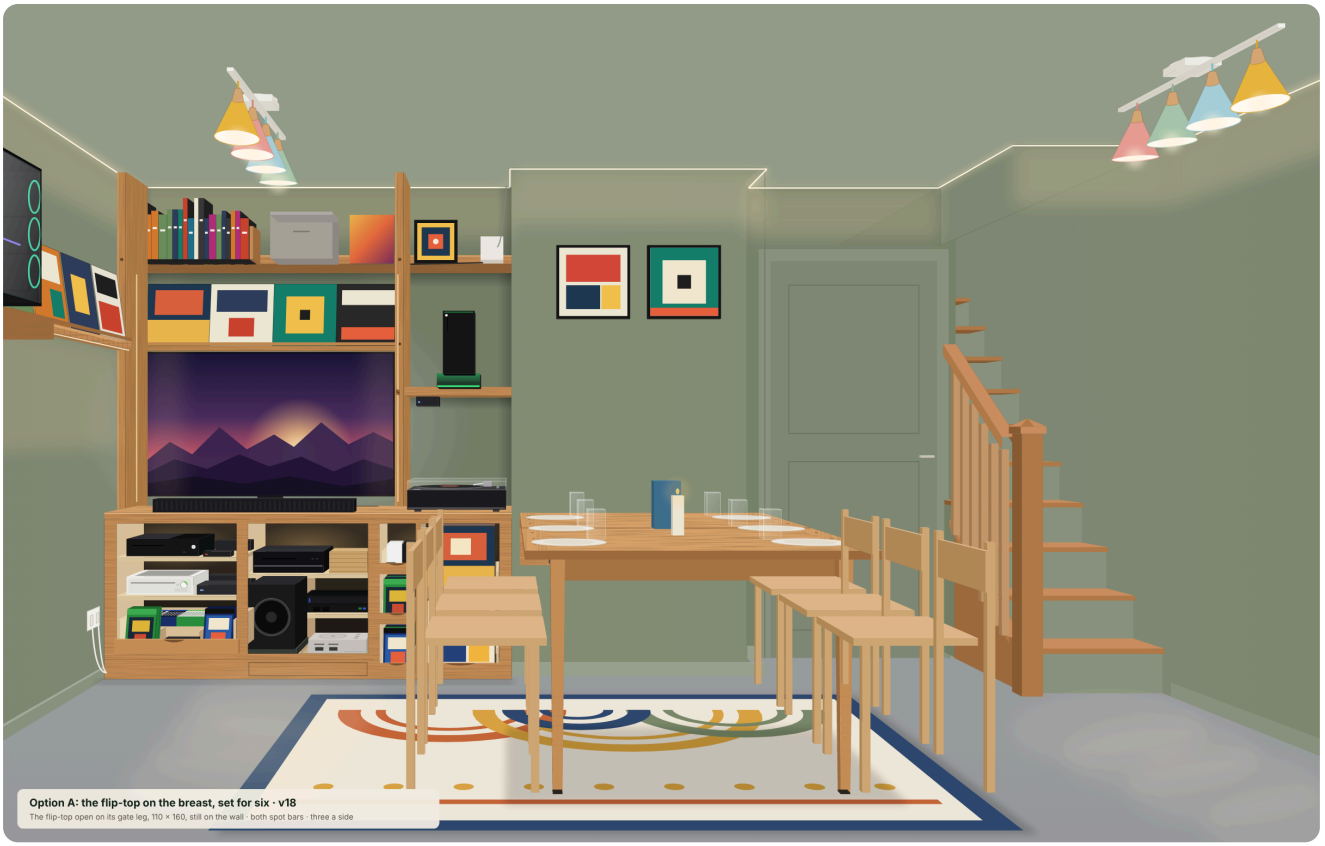
Total		**About 191 LPs, every one face-out**

If the collection is bigger than this, tell me the count. The neatest overflow is another LP column like bay 3's (40 wide, two crates, 180 LPs), or LPs spine-out on the high shelf.

5 · The dining wall

v16: the table by the stairs (option B) is now the plan's default, as you asked; the chimney breast (option A) stays drawn, checked and costed as the alternative. The dining table is a **flip-top gate-leg table that never leaves the wall** (new in v15: you asked for one that stays attached and seats more). A solid oak top, 110 × 80 cm, hangs on lift-off hinges on an oak backboard and rests on two folding brackets, as before. A second 110 × 80 oak half is hinged to its front edge on three brass flip-top hinges[30]: flip it out and the table runs 160 cm into the room, its far end on a folding oak gate leg, at the same 76 cm as the worktop. Four sit with room to spare, six elbows-in; by the stairs a seventh sits at its end. Every day it folds flat against the wall, about 13.3 cm deep, the half and its gate tucked behind the top. Nothing comes off the wall, there's no leaf to store, and no legs to screw in.

Option A, open on the breast and set for four: the top on its brackets, the half flipped out beyond it on its gate leg, two down each side. Bar 2, on the pendant point by the stairs just out of view, lights the table and the framed LPs.

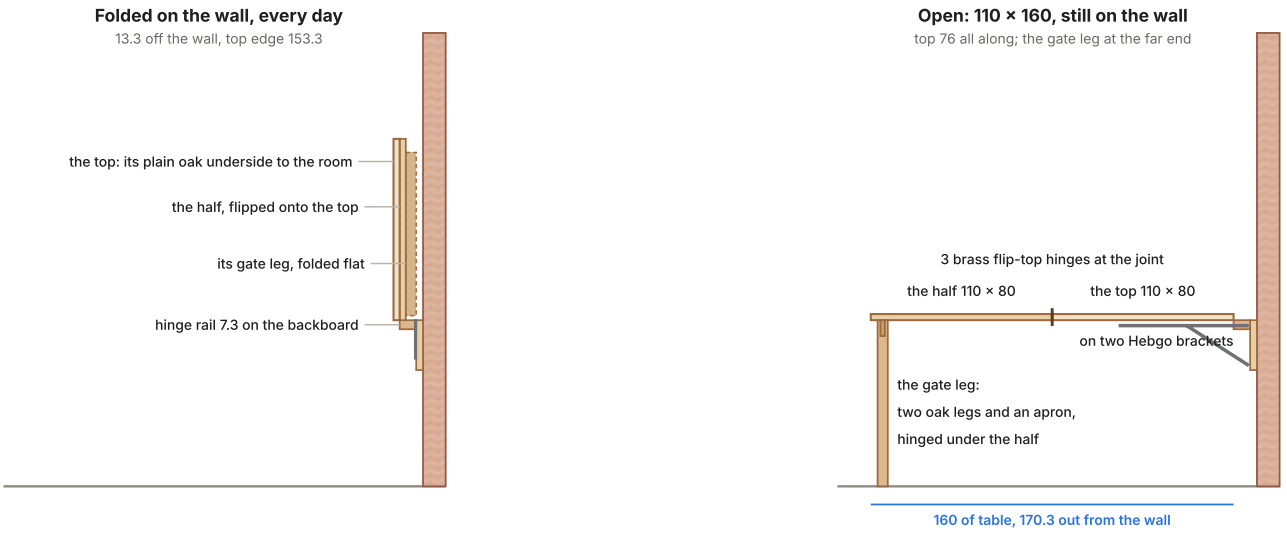


Set for six, three down each side (53.33 cm each: elbows in). The camera stands back in line with the alcove's edge, so both spot bars show. The table's end stops 30.7 cm short of the sofa, so on the breast nobody sits at its end.

The flip-top table: folded, and open

In section through its centre line, to scale, the wall on the right of each. It never comes off the wall, and there's no leaf to store.

1. To open: turn the two buttons and lower the pair onto the brackets as they swing out and lock (two people: about 37 kg).
2. Flip the half over onto the room side: it opens on its hinges to lie flush with the top. Swing the gate leg down until it stops.
3. To fold: gate leg up flat, half back over onto the top, then lift the pair up to the wall and turn the buttons.



Two places for it: under your shelf by the stairs (the default), or the chimney breast

You asked (v14) what it would look like on the wall going up the stairs, under your 2 m shelf. Your photos put that wall at about 268 cm from the stair box to the window corner, with the shelf about 196 cm up. The flip-top works on

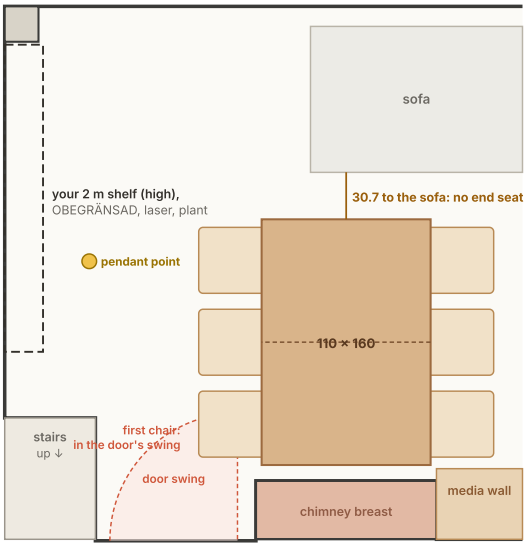
either wall: only the backboard's fixings change. Both are drawn in the 3D room on the plan page, folded and open, from every angle and in each lighting scene.

Where the dining table can live: the chimney breast or the stairs wall

Both to scale, the flip-top open at its fullest, the west half of the room from above (window at the top). It never comes off the wall: four, six or seven sit at the same tab

A · On the chimney breast: up to six

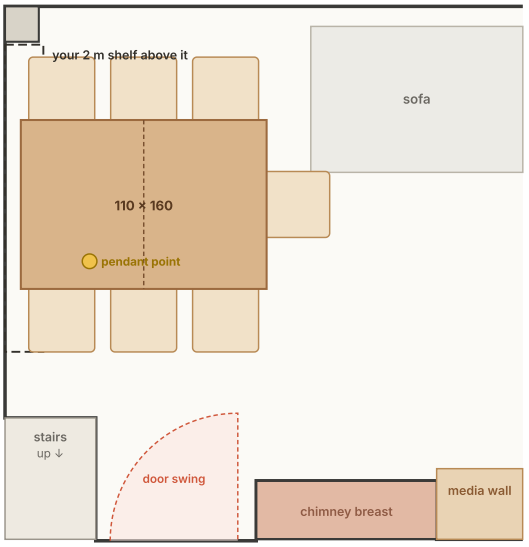
the flip-top open towards the window, three a side; folded flat every day



+ next to the media wall; the stairs wall keeps the shelf, lamp and hat
+ four sit with room to spare, six elbows-in
− no seat at the end: the sofa is 34 beyond it
− the west side's first chair sits in the door's swing

B · Under your shelf, by the stairs: up to seven (the default)

the flip-top open into the room, three a side and one at the end



+ seven at a squeeze, five with room; right under the pendant point
+ out of the door's swing; frees the breast for the “now playing” ledge
− the side chairs are tight: 43 to the stairs' foot, 33 to the window
− the lamp moves while it's open; check the wall before fixing the brackets

Option B in the 3D room, set for dinner: the flip-top open into the room under your shelf, bar 2 on the existing pendant point over it, the stairs behind the first chairs.

Option B folded on an everyday evening: 13.3 cm deep against the wall, 37.7 cm under the shelf with your black OBEGRÄNSAD panel, the laser (off) and the plant hanging from the window end beside the pipe boxing; your red lamp in the corner.

- **A, on the breast (the alternative): up to six.** It opens towards the window, 170.3 cm out, three a side. Its end stops 30.7 cm short of the sofa, so there's no seat at the end; for six, slide the coffee table towards the TV first; the west side's first chair sits in the door's swing, so keep the door shut while you eat (or rehang it to open into the hall).
- **B, by the stairs (the default): up to seven.** It opens into the room from the stairs wall, three a side and one at the end, right under the existing pendant point and out of the door's swing; it frees the chimney breast. The side chairs are tight (43 cm to the stair box behind the stair side, 33 to the window wall), so tuck them in after a meal; your red lamp moves from the window corner while it's open. The west wall's build is unknown: knock on it before choosing (solid takes plugs; studwork means fixing the backboard into the studs).
- **What moves for B:** the dresser goes (its turntable and records already have homes in the media wall, all face-out); the sofa stays where it is.

How it works

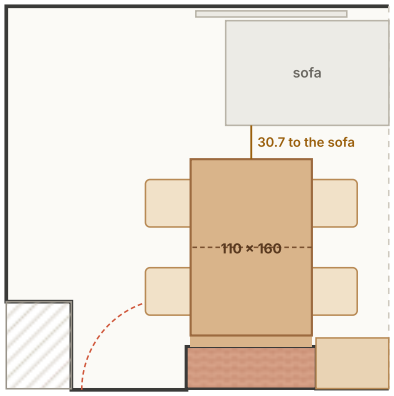
- **The top and the half:** solid oak with **pegged breadboard ends**, 7 cm oak strips across each end on a tongue, pinned with three oak pegs in slotted holes so the boards can move. They stop the panels cupping when they live upright, and they frame them like the uprights frame the TV. Rounded 5 cm corners on the half's free end, 1 cm chamfers underneath so it reads slim.
- **Backboard and hinge rail:** oak, 110 × 22 × 3 from the worktop's offcut, screwed to the wall with plugs every 20 cm; along its top, a 110 × 4 × 7.3 oak hinge rail stands it out so the folded half and its gate fit behind the top. The rail's top is flush with the open table, at 73.3 under the top.
- **Brackets:** two Häfele Hebgo heavy-duty folding brackets (287.43.428: 580 mm arm, 500 kg a pair[18]), 80 cm apart on the backboard's face, under the top. Raised, they lock on their own; a light lift on the arm releases them.
- **Hinges:** three 100 mm lift-off hinges[19] along the rail's front edge carry the top; three brass flip-top hinges, 1.27 × 7 cm, set into the meeting edges of the top and the half, open to 180°, so the half lies flat on the top to fold and flush with it open.
- **The gate leg:** two tapered oak legs (4.5 to 3 cm, 73.3 long[20]) joined by a 7 cm oak apron, hinged under the half 3 cm in from its far edge on two butt hinges, with a folding stay each side to lock it upright. It folds flat inside the half, along its 80 cm.

- **Folded:** two oak turn buttons on the wall, plus a magnetic catch, hold the top edge at 153.3. The top's underside faces the room: plain oiled oak between the pegged breadboard ends, so folded up it reads as a calm oak panel under the two framed LPs. It's about 37 kg to swing up or down: two people, one at each corner.
- **Chairs:** drawn at IKEA TERJE size (43 × 49, seat 48[22]), which folds flat. Any dining chair with a 45–48 cm seat works.

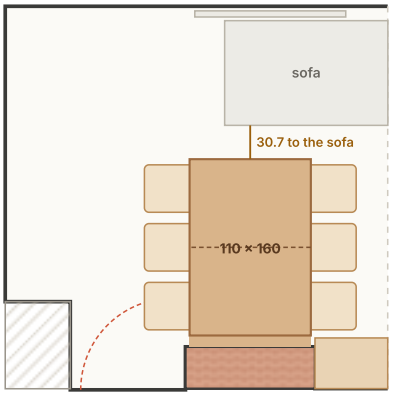
The flip-top table for four, six or seven

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter

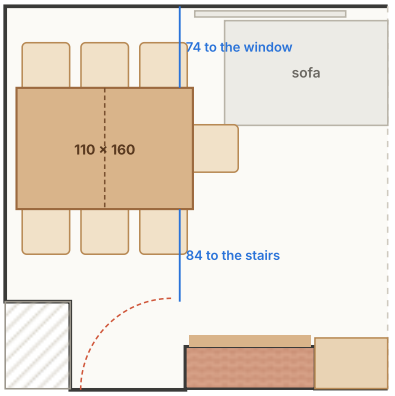
Four, on the breast
the flip-top open: two down each side



Six, on the breast
three down each side, elbows in



Seven, by the stairs
three a side and one at the end



• Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
• The flip-top never leaves the wall; the dashed line is the joint where the half flips out on its gate leg. Red: the door's swing.
• Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

Seats	Where	Per diner	Clearances
Four	Either wall, open: two down each side	80 each	On the breast its end is 30.7 from the sofa; by the stairs the side chairs tuck in
Six	Either wall, open: three down each side	53.33 each (55 is a squeeze)	On the breast slide the coffee table towards the TV and keep the door shut
Seven	By the stairs only: three a side and one at the end	53.33 each; the end has the full 110	The end chair clears the sofa's front; the lamp moves

Clearances from a UK planning guide: 60 cm of table edge per diner, 60 cm from the table to a wall as a tight minimum, 75 to aim for, 100–110 to walk behind someone seated[23].

Cut list and hardware

Part	Material	Size (cm)	Qty
Table top · the half	Solid oak, edge-glued staves	96 × 80 × 2.7 centre + two 7 × 80 breadboard ends (110 overall), each	1 · 1
Breadboard pegs	10 mm oak dowel, slotted holes in the tongue	3 per end	12
Gate legs · apron	Solid oak, the legs tapered on two inside faces	73.3 × 4.5 tapering to 3 · 94 × 7 × 2	2 · 1
Backboard · hinge rail	Solid oak, from the worktop offcut	110 × 22 × 3 · 110 × 4 × 7.3	1 · 1
Turn buttons	Oak offcut	6 × 2.5 × 2.2, on a single screw	2
Folding brackets	Häfele Hebgo 287.43.428	arm 58, height 22	2

Lift-off hinges	Union PowerLoad, 100 × 88	one 3-pack, same hand	3
Flip-top hinges	Lee Valley 40574, solid brass	1.27 × 7	3
Gate hinges · stays	Steel butt hinges, 75 mm · folding (locking) stays	—	2 · 2

Safety: never fold it up with anything on it, keep both turn buttons turned when it's up, check the brackets have locked (a click each side) and the gate's stays are locked before you load it.

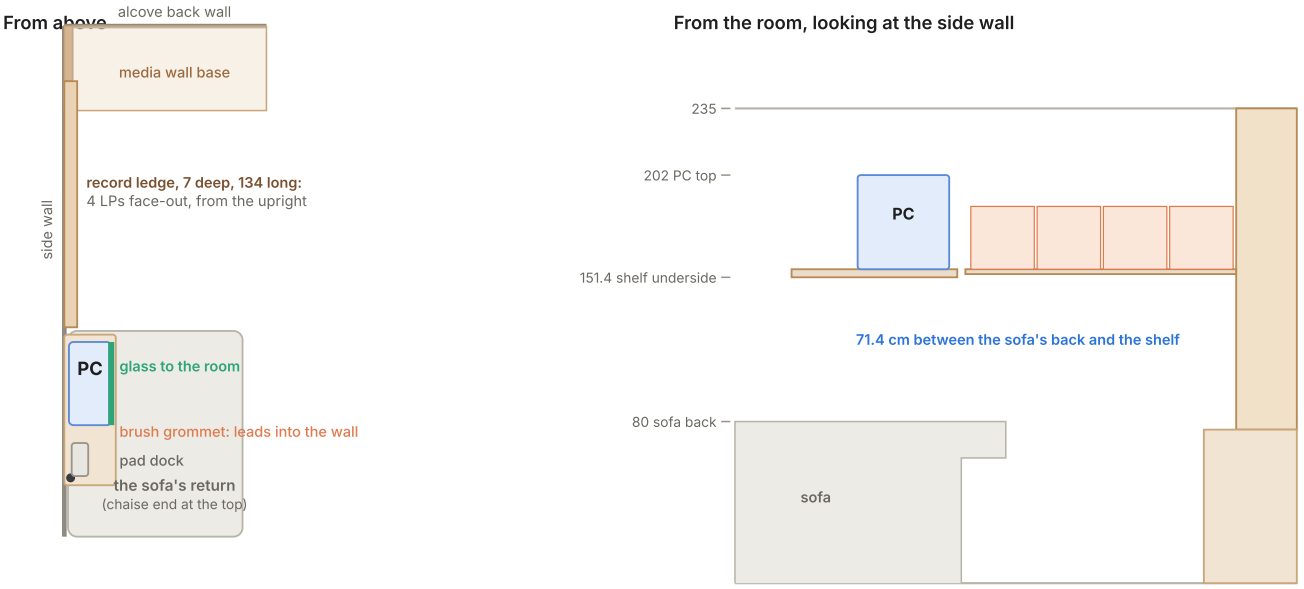
6 · The side wall

One oak line at 155.4 cm runs from the ledge over the TV, round the corner on the left upright, and along the side wall. New in v15, the side wall's part is **one run**, 4 cm thick all along: a 7 cm record ledge with four more LPs face-out that **sweeps out in one smooth curve** to the 28 cm shelf the PC sits on, ending in a 5 cm radius like the table's corners. A warm LED strip in a groove under its front edge washes the wall below it, on the uprights' driver and dimmer. Two separate shelves at two depths become one piece of furniture, and it moves the PC out of the TV's sightlines.

The side wall from the room: the oak run round the corner with its four LPs, sweeping out under the PC, the keyboard tray's housing underneath and the LED's glow below it, the curtains at the window, and nothing else on the wall: the leads run inside it, the sofa below (drawn from your photo), and the media wall end-on at the right. The chimney breast is cut away so the wall reads end-on; its outline is dashed, with the depths marked: the alcove is 38 deep, so 38 of the 46 cm base sits inside it and 8 stands proud (picked out in gold), and the 30 cm uprights and shelves sit wholly inside.

The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines



- **The run:** 220 cm long, 4 thick, cut from one 225 × 30 × 4 oak board to a full-size template: 7 deep along the LPs (with a 1.5 cm lip), easing out to 28 over 14 cm just past the last LP. It sits on an oak cleat screwed to the side wall, dowelled to the left upright's front edge.
- **The PC end:** 82 × 28 cm, on two concealed heavy-duty brackets (40 kg+ a pair) as well as the cleat, 71.4 cm above the sofa's back. The case sits with its glass to the room and its front towards the alcove. The ports face the window and stay open to the air (v13: no screen, so nothing traps the exhaust); from the sofa you look at the glass side, not the ports. The pad dock sits beyond the case.
- **Cables:** through a brush grommet at the shelf's back corner straight into a 25 mm conduit chased into the side wall (it's solid, as you confirmed), down to skirting height and behind the skirting to the base's plinth; you chase it yourselves in step 3, with a second, smaller conduit beside it for the PC's mains lead, then it's filled and painted, so nothing shows. A draw wire left in it pulls the next lead through. The HDMI is an active optical cable (about 5 m), "source" end at the PC.
- **Power:** the PC plugs into the 10-gang behind the TV: its lead (a 5 m C13 lead with a rewireable plug, fed down bare and the plug fitted at the bottom) runs in its own 25 mm conduit beside the data one. The 10-gang's load check, with the PC, is in section 9. No new socket.
- **Keyboard tray (back from v6.1):** the shelf sits on a 7 cm oak housing with a tray on 300 mm light-duty runners that rolls 25 cm out towards the room. It holds a full-size wireless keyboard and mouse: pull it out, lift the keyboard onto your lap, park it again. The housing stops 12 cm short of the sofa end so the cables drop past it; its underside is 144.4, 64.4 above the sofa's back.
- **Headroom:** the shelf's underside is at 151.4, and it covers only the 28 cm of the sofa nearest the wall.

The PC end, looked over again (v17)

- **Air:** a mid-tower draws in at the front (towards the alcove) and blows out at the back and top. Both stay open: the case's top is at 202, 33 clear of the ceiling; a magnetic dust filter on the top exhaust keeps the dust out while it's off.
- **Reach from the sofa:** the PC's own power button is on its top, behind your head. A remote power button (a small card on the motherboard's power pins and a push button under the wrap ledge by the sofa end) starts it from the seat, and a powered USB 3 hub under the run's front edge puts pads, the keyboard's dongle and a stick within reach.
- **Lit, and seen:** the glass side faces the room, so its fans and lights read as part of the wall at night (the 3D view's "PC lit" button shows it). Keep its lights on a slow single colour if they pull the eye from the TV.
- **Leads:** labelled at both ends (step 13), all three down the chased conduits; nothing hangs below the shelf.

7 · The oak joinery

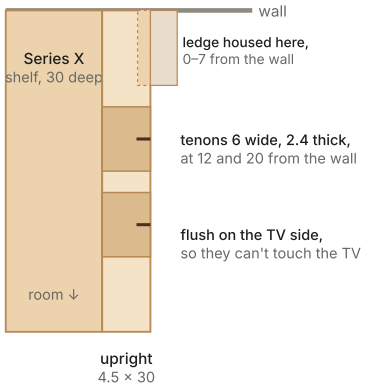
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall showing between them.

- **Uprights:** solid oak, 4.5 × 30 cm, 159 cm long, each on two 10 mm dowels into the worktop, scribed to the ceiling and held to the back wall with concealed brackets; the left one also screws to the side wall.
- **Wedged through-tenons:** the Series X shelf (top 134) passes through the right upright on two tenons, 6 cm wide and 2.4 thick, 12 and 20 cm from the wall. They're wedged and planed flush: standing proud, they'd touch the TV.
- **Housed and pegged:** the high shelf (two lengths, top 196) and the record ledge (top 155.4) are housed 1.2 cm into the uprights and pinned with 10 mm oak pegs, which show as dark dots.
- **LED grooves:** a routed groove on each upright's inner face holds a smart colour strip (v16) that washes the wall either side of the TV: warm white day to day, any colour for a scene.
- **Hidden cables:** a 2 × 1.5 cm groove down the back of the right upright carries Deco B's 12 V lead and patch lead, the Series X's HDMI and power, and the vinyl lead.
- **Finish:** hardwax oil throughout; lightly limed or smoked oak avoids the bright orange "new oak" look.

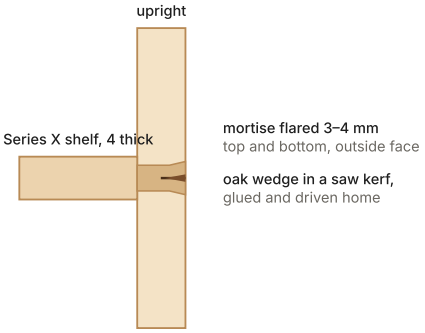
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

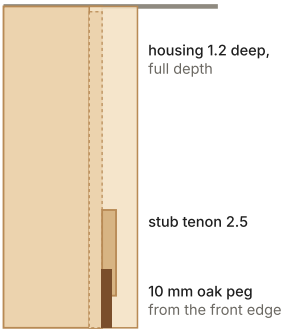


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

• Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

Joinery cut list

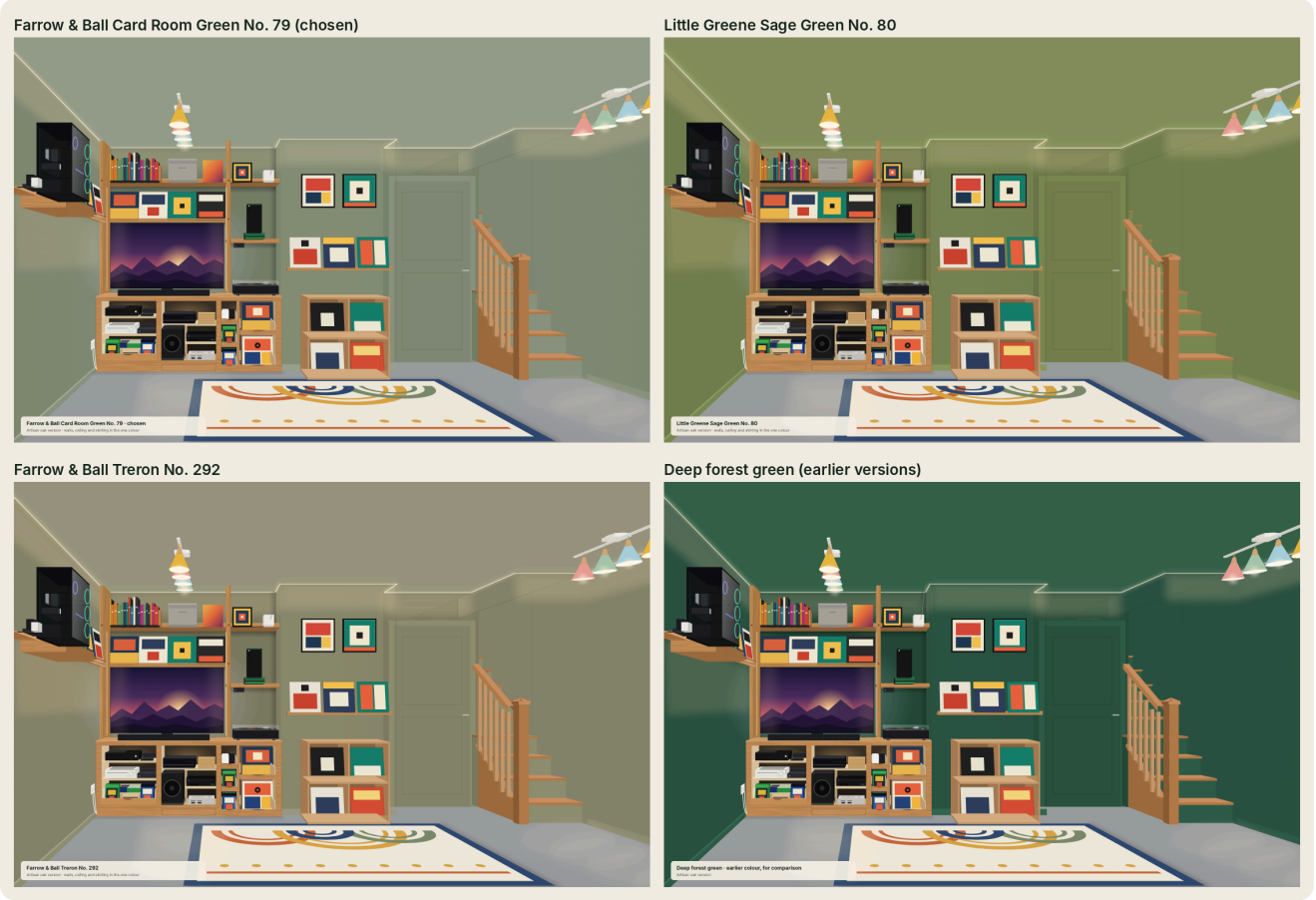
Part	Material	Finished size (cm)	Joints	Qty
Uprights	Solid oak, quarter-sawn if you can	159 × 30 × 4.5 (buy 165 and scribe)	Housings for the high shelf and ledge; mortises for the Series X shelf in the right one; LED groove inside; cable groove at the back of the right one	2
High shelf	Solid oak	128.9 × 30 × 4 and 51.7 × 30 × 4	Housed 1.2 into the uprights, pegged; the short length on a cleat on the reveal	2
Record ledge	Solid oak	128.9 × 7 × 2.4; lip 0.9 × 1.5	Housed and pegged	1
Series X shelf	Solid oak	55 × 30 × 4 (50.5 + 4.5 through)	Two through-tenons 6 × 2.4, wedged flush; cleat on the reveal	1

Side-wall run (v15)	Solid oak, one 225 × 30 × 4 board	220 long, 7 deep sweeping to 28, 4 thick; lip 0.9 × 1.5 along the 7 cm part	Cleat, two concealed brackets under the PC end, two dowels to the left upright; LED groove under the front edge	1
Worktop and backboard	One solid oak worktop, 300 × 62	186 × 46 × 3 and 110 × 22 × 3	Chamfered fronts; cut edges sealed	1 board
Face frame	Oak, 20 mm	186 × 73, stiles about 5 wide	Glued and pinned; end stiles scribed	1 set
Wedges · pegs · dowels	Oak · 10 mm oak dowel	6 × 3, tapering · 5 long	Glued	2 + spares · 6 · 8
Keyboard tray	Oak housing sides 7 × 2, ply tray, oak lip	70 × 28 × 7; tray 54 × 26	Housing screwed under the shelf; 300 mm runners	1

8 · Paint and light

Paint: Farrow & Ball Card Room Green No. 79

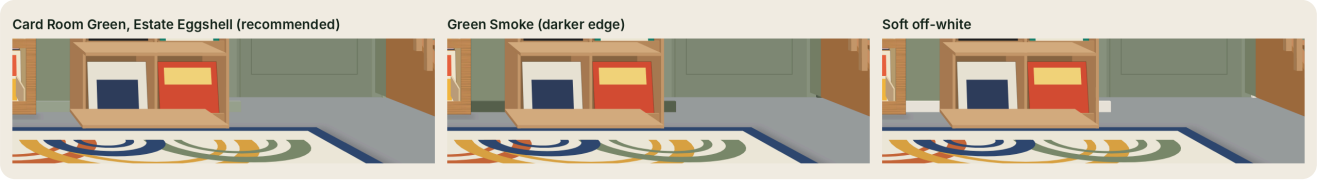
The room is colour-drenched in Card Room Green: walls, ceiling, alcove, chimney breast, skirting, door and frame and the radiator. There's no trunking left to paint. It's a soft grey-green that lets the oak and the records lead.



The wall in Card Room Green (chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening: judge from painted samples.

- **Finishes:** Estate or Modern Emulsion on walls, ceiling, alcove and breast; Estate Eggshell in the same colour on skirting, door, frame and radiator (Modern Eggshell if the skirting takes knocks).
- **Order:** paint before anything is fixed, so nothing needs cutting in round the uprights, shelves or the dining wall.

Skirting: Card Room Green too



Recommended: Card Room Green in Estate Eggshell, which completes the drench and hides scuffs far better than white. Darker option: Green Smoke. Over the white gloss: wash, key, woodwork primer, two coats of eggshell.

Light, in three warm layers

- **The LED strip on the ceiling** (as now): round the room's edge, into the alcove and across the breast; 2700K on a dimmer.
- **Built into the oak (v16: smart colour):** an RGB and tunable-white strip in each upright's groove and one under the side-wall run's front edge[37], in the same app as the bars' bulbs, so each scene sets them too; on their plug-

in power on the 10-gang (no fused spur). The 6500K bias light on the back of the TV stays plain white: it's there for the picture, not the mood.

- **Your red lamp:** the three-headed floor lamp you have, its colour bulbs green at the top, pink in the middle and purple at the bottom. A deep green room needs a low layer as well as the light from above; it's also the one to leave on for a film. It can stand in any of the three spots in the 3D room.
- **Two spot bars** in the style of Dunelm's Wolston (£49 each; 84 × 16 cm, 22.4 cm drop; 4 × E14, 6 W max[14]) on your two existing ceiling points, **both running from the TV towards the window** (v15). Bar 1 hangs on the rose straight out from the TV unit, halfway to the sofa; bar 2 on the pendant point by the stairs, along the west wall, so its end no longer comes near that wall. **Smart colour bulbs** (v15): eight E14 bulbs under the bar's 6 W, the WiZ Colour P45 (4.9 W, 4.5 × 10.1 cm) or, if it shows below a shade, the shorter Philips Hue lustre (5.1 W, 4.5 × 7.7 cm)[38]. Smart bulbs need full power, so **no dimmers**: the wall switches stay as they are and stay on; level and colour come from the app (or an optional smart button).

Where the heads point (v17): all eight heads straight down, evenly spread along each bar, so their pools overlap into one even light over the room: that's the **General** scene, the default. Tilt a head only if its pool lands badly (in the TV's screen from the sofa, or in someone's eyes). Set the bulbs to warm white (about 2700K) for General, and save colours for the scenes you want them in.

Lighting scenes (v13.7; smart since v15)

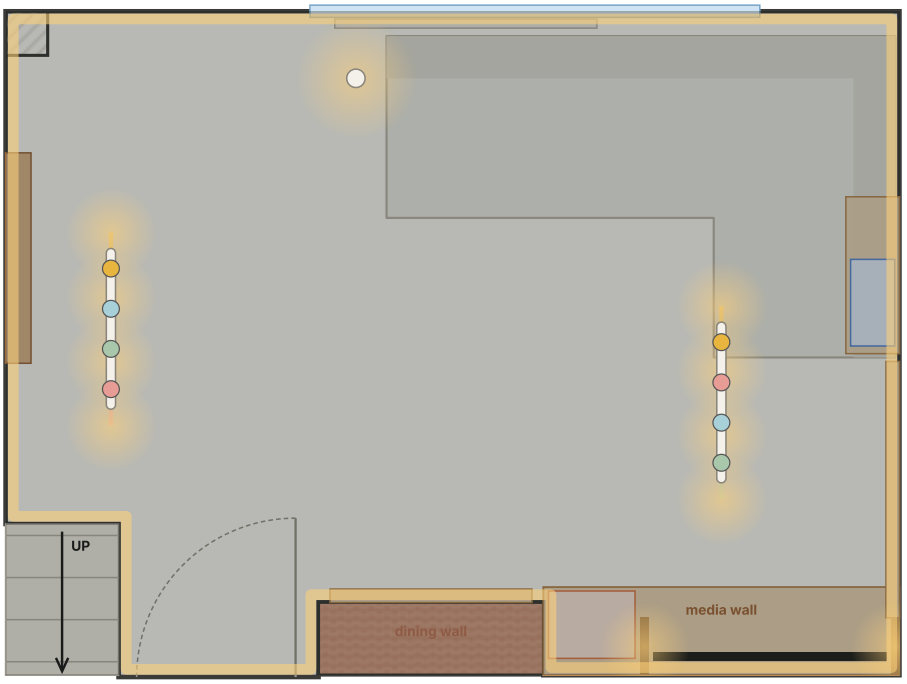
Five ways to set the room (v17: General, the default, first): the strip's controller, the inline dimmer on the uprights' LED driver, and the bars' and the lamp's smart bulbs in their app. Save these four as scenes in the app (and on the smart button if you get one). In the 3D room each light has a slider, the bars and the lamp a colour, and the scene buttons fill the sliders in.

Light	Control	General	TV night	Dinner	Records	Everyday evening
Ceiling LED strip	its own controller	55%	10%	25%	35%	70%
Uprights' grooves and the side shelf	the strips' app (smart colour since v16)	45%	15%	30%	60%	50%
TV bias light	comes on with the TV	off	on	off	off	off
Bar 1, between the TV and the sofa	smart bulbs: the app	60%	off	15%	60%	55%
Bar 2, by the stairs	smart bulbs: the app	60%	10%	80%	15%	40%
Your red lamp	its colour bulbs: the app	45%	70%	off	80%	60%

Levels are brightness in the app for the smart bars and the lamp, and roughly how far up the strip's controller and the uprights' inline dimmer go. A scene changes how bright the heads are (and their colour), not where they point.

Lighting scene: General

An even light over the whole room: every head straight down, the strips and the lamp filling in. The default (v17).



Ceiling LED strip
55% · its own controller

Uprights' grooves and the side shelf
45% · the strips' app (smart colour since v16)

TV bias light
off · comes on with the TV

Bar 1
60% · smart bulbs: the app

Bar 2, by the stairs
60% · smart bulbs: the app

Your red lamp
45% · its colour bulbs: the app

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- All eight heads point straight down, spread a little along each bar, so the pools overlap instead of spotlighting the walls.
- The ceiling strip and the uprights fill the edges; the lamp adds a low warm layer.
- Tilt a head only if its pool lands badly (on the TV's screen, or in someone's eyes).

Lighting scene: TV night

Films and games: the screen is the brightest thing in the room, with a little warm light round it so your eyes don't strain.

Ceiling LED strip
10% · its own controller

Uprights' grooves and the side shelf
15% · the strips' app (smart colour since v16)

TV bias light
on · comes on with the TV

Bar 1
off · smart bulbs: the app

Bar 2, by the stairs
10% · smart bulbs: the app

Your red lamp
70% · its colour bulbs: the app

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1 off: its heads hang in front of the screen, and even tilted away they'd catch it.

- The bias light behind the TV and your red lamp do the work: the lamp is the one to leave on for a film (down to about 70).
- Bar 2 just glowing on the framed LPs, so the far end of the room isn't a black hole.

Lighting scene: Dinner

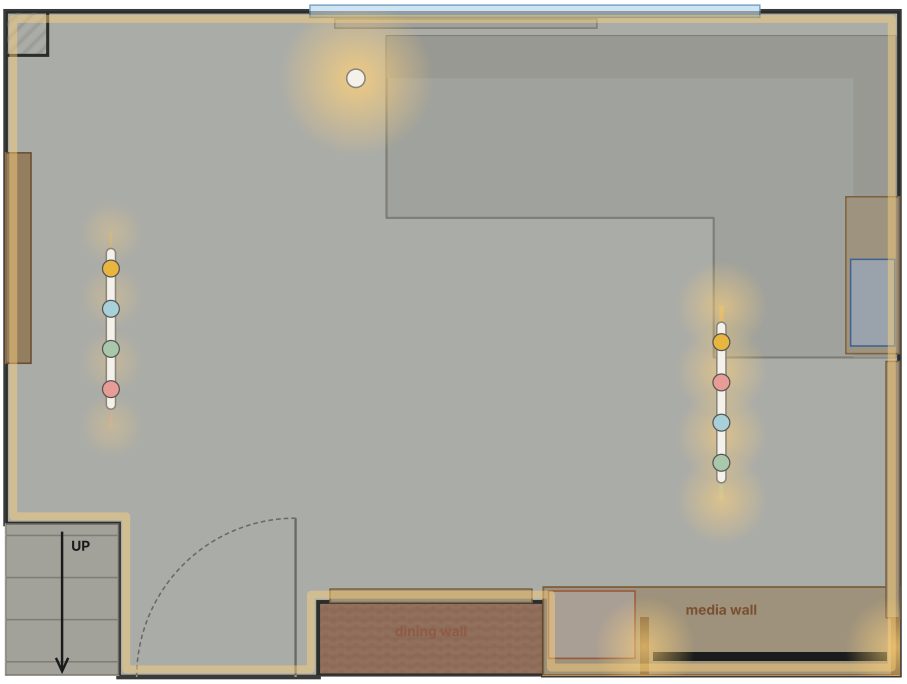
The table is the bright spot, the media wall steps back.

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 2 up: for four its heads light the table at the wall; for six, the table stands right under it (point blue and mustard straight down).
- Bar 1 low, so the coffee table end isn't dark; the TV off.
- The uprights low: the shelves read as a backdrop, not a stage.

Lighting scene: Records

Pick a record, read the sleeve: light on the deck, the ledge and the trays.



Ceiling LED strip
35% · its own controller

Uprights' grooves and the side shelf
60% · the strips' app (smart colour since v16)

TV bias light
off · comes on with the TV

Bar 1
60% · smart bulbs: the app

Bar 2, by the stairs
15% · smart bulbs: the app

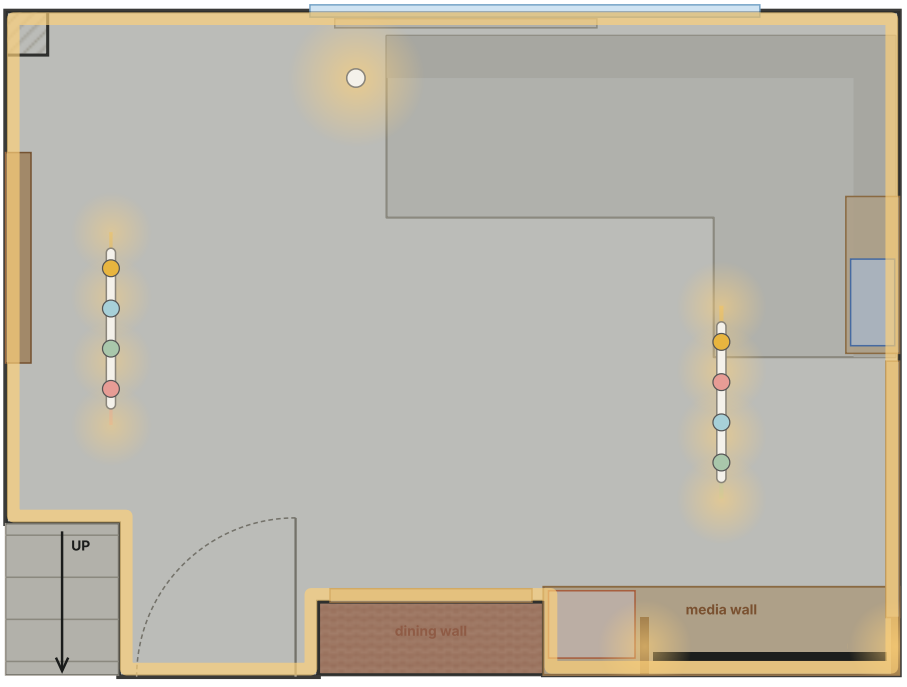
Your red lamp
80% · its colour bulbs: the app

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1 up: its pink head is on the turntable and the Series X, blue on the record ledge.
- The uprights' grooves up, washing the wall either side of the (switched-off) TV and the ledge LPs.
- The lamp up for reading sleeves on the sofa.

Lighting scene: Everyday evening

Everything on at a comfortable level, for tidying, chatting and finding the remote.



Ceiling LED strip
70% · its own controller

Uprights' grooves and the side shelf
50% · the strips' app (smart colour since v16)

TV bias light
off · comes on with the TV

Bar 1
55% · smart bulbs: the app

Bar 2, by the stairs
40% · smart bulbs: the app

Your red lamp
60% · its colour bulbs: the app

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- The ceiling strip does most of it; both bars fill in the shelves and the table.

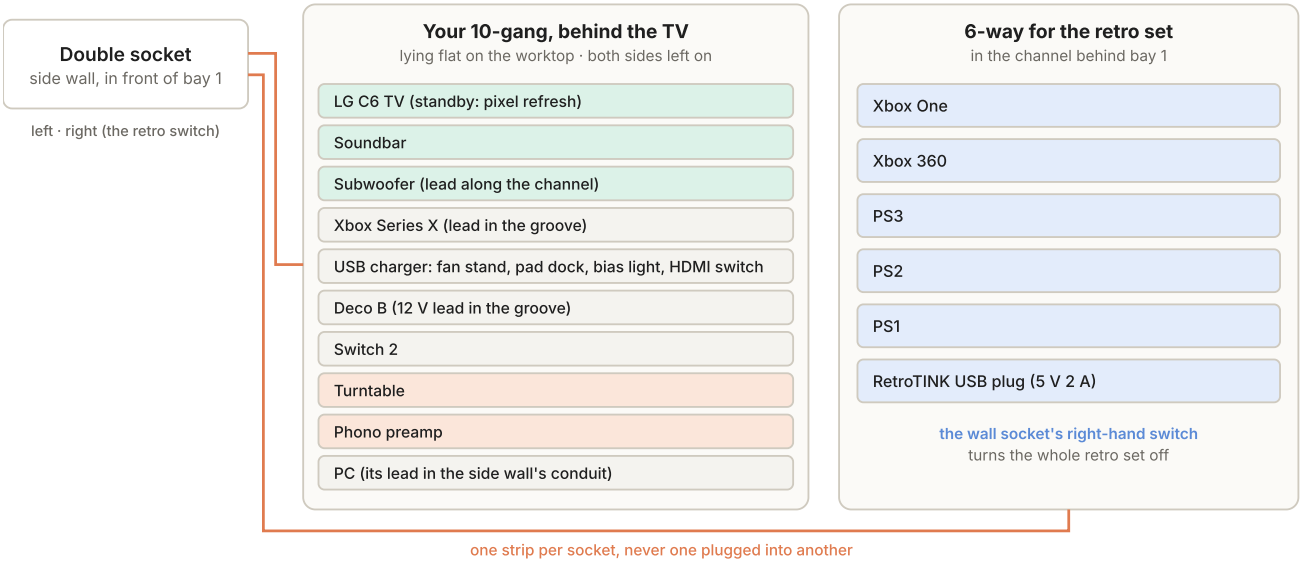
- Turn the strip down first as the evening goes on: it's the flattest light in the room.

9 · Power, cables and network

Power

Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: step 3).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: the socket by the sofa, if there is one. If not, a 12-way surge-protected strip in place of the 10-gang gives them a way.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

- **The socket:** the single on the side wall just in front of bay 1, about 30 cm up, becomes a double at the same spot. The left outlet feeds the 10-gang; the right one feeds the retro 6-way, so its own switch is now the retro switch. The 10-gang's switches sit behind the TV, 2 cm from the corner post, where you can't reach them any more.
- **10-gang:** lying flat on the worktop behind the TV, both sides left on: TV, soundbar, sub, Series X, USB charger, Deco B, Switch 2, the deck, the preamp and now the PC. With everything running that's about 1.55 kW (the PC's 850 W at most), under the strip's 13 A (3 kW).
- **Retro 6-way:** in the channel behind bay 1 (bay 2's back is the sub's port): Xbox One, Xbox 360, PS3, PS2, PS1, RetroTINK.
- **OLED care:** leave the TV in standby so it runs its pixel refresh.

Socket by socket

Socket	Feeds	On or off
Side wall, double, left outlet	The 10-gang behind the TV: TV, soundbar, sub, Series X, USB charger (fan stand, pad dock, bias light, HDMI switch), Deco B, Switch 2, the turntable, the preamp, the PC and the uprights' LED driver	Always on: about 1.55 kW with everything running, PC included
Side wall, double, right outlet	The retro 6-way in the channel: Xbox One, Xbox 360, PS3, PS2, PS1, the RetroTINK's USB plug	The wall switch is the retro switch: off between sessions; up to about 0.8 kW with a launch PS3

By the sofa, if there is one	The rear speakers (if there's no socket there, a 12-way strip in place of the 10-gang gives them a way)	On
Ceiling points 1 and 2	The two spot bars, with smart colour bulbs (v15)	Switches left on; level and colour from the app

One strip per outlet, never one plugged into another, and each under its 13 A (3 kW) limit: about 1.55 kW on the 10-gang and up to 0.8 kW on the retro 6-way, about 10 A together.

Electrics: your own job

You're doing the electrics yourselves. In England, Part P of the Building Regulations makes only three kinds of work notifiable: a new circuit, a replacement consumer unit, and any addition or alteration to a circuit in a special location (round a bath or shower). Replacing an accessory like for like, and adding to an existing circuit anywhere else, isn't notifiable, though it must still be done to the wiring regulations (BS 7671). So the jobs here need no electrician and no building control: the socket swapped for a double, and the two spot bars in place of the old fittings (the light switches stay: the bars' smart bulbs need full power, so there are no dimmers to fit). The Cat6 and the conduits aren't mains work at all.

The safe routine, every time

1. **Isolate:** switch off that circuit's breaker at the consumer unit (the lighting circuit for the bars, the socket circuit for the socket). If the labels aren't clear, switch off the main switch.
2. **Lock off:** fit the MCB lock-off and padlock, keep the key in your pocket, and tape a note on the board.
3. **Prove dead:** check the tester lights on its proving unit, test between every pair of terminals (live–neutral, live–earth, neutral–earth), then check the tester on the proving unit again.
4. **Photograph** the wiring before you disconnect anything, and connect the new one wire for wire: brown (or red) to L, blue (or black) to N, green-and-yellow to earth, sleeved, no copper showing, every screw tight.
5. **Test:** power back on, then the socket tester in both outlets; each bar's four bulbs come on and join the app.

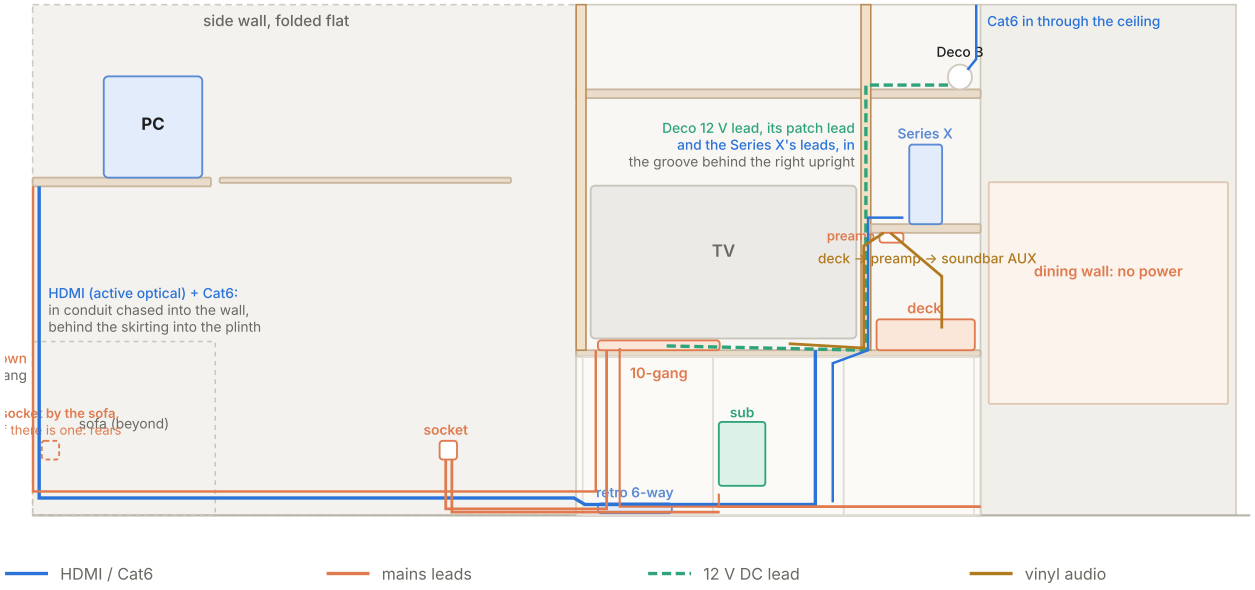
Stop and call an electrician if…

- the tester shows anything live after you've isolated;
- the wiring is old (rubber or fabric sheath, or red and black with no earth), or there's no earth at a ceiling point (the spot bars need one);
- there are more wires than you can match to your photo (two-way switching, a loop in a ceiling rose you can't follow), or scorch marks, or a loose, hot or crumbling fitting;
- the consumer unit has rewireable fuses or no RCD;
- a job grows into a new circuit, consumer unit work, or anything by the bath or shower.

Cable routes

Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits in bay 2

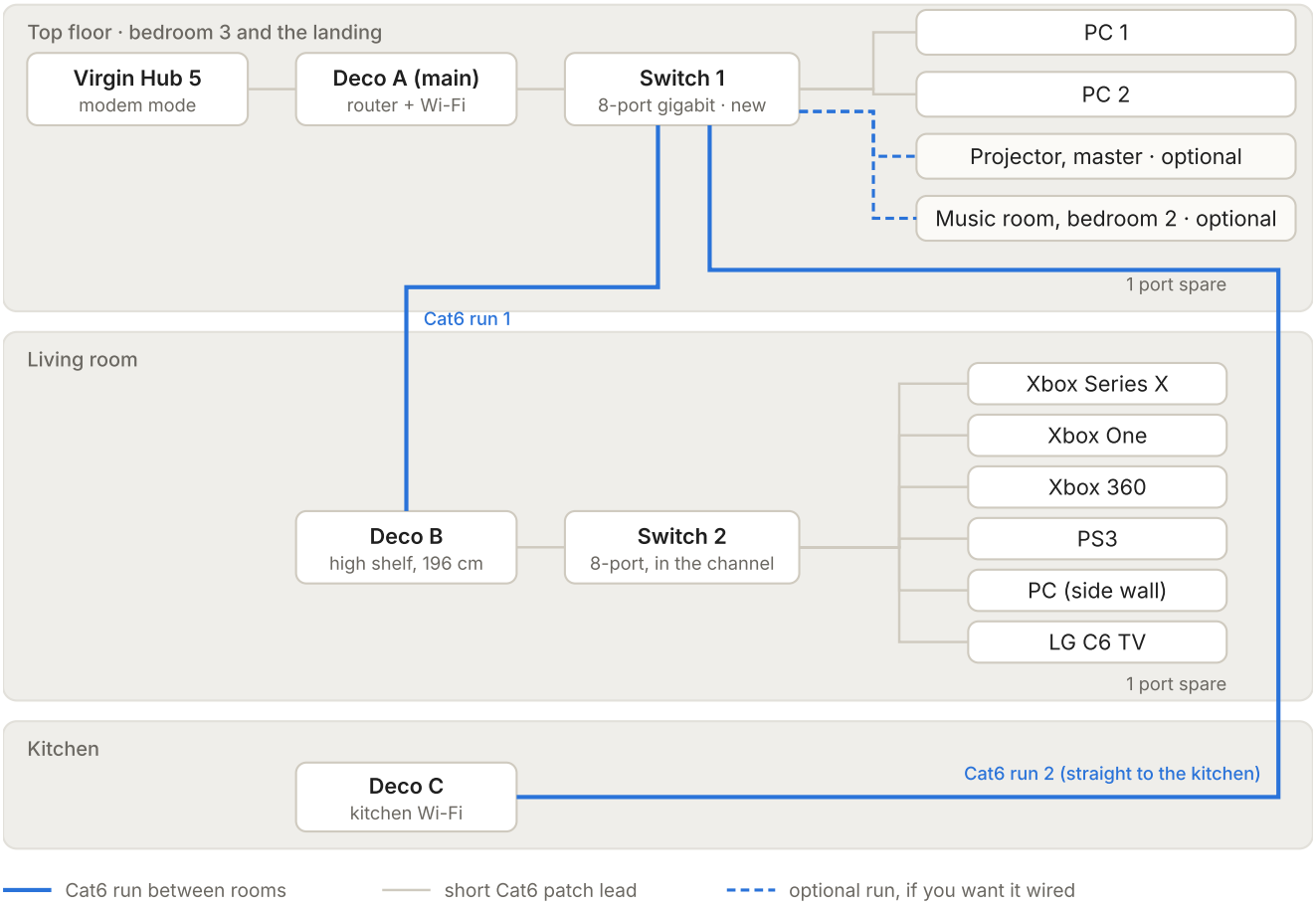


- **The channel:** 4 cm behind the frames, flat two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Switch 2 and the retro 6-way stand on edge in it.
- **The groove behind the right upright:** Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead, down to the worktop's slot.
- **The PC:** through the shelf's brush grommet into two conduits chased into the side wall (the HDMI and Cat6 in a 38 mm one, its mains lead in a 25 mm one), behind the skirting and into the plinth notch; the mains lead goes on to the 10-gang.
- **The sub:** its mains lead from the 10-gang drops through the slot behind the TV to the back of bay 2, about 1 m. Keep the channel behind the sub clear: its port needs the air.
- **Airflow:** console bays stay open-backed; the Series X has 15 cm each side and 21 above.

Network

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



- **Bedroom 3:** the Hub 5 (in modem mode), Deco A and Switch 1 sit together with the gaming PCs. Switch 1's eight ports: the uplink, two PCs, the two Cat6 runs down to Deco B and Deco C, and, if you want them, the projector in the master and the music room in bedroom 2.
- **The route down:** across the landing to the airing cupboard, which sits over the alcove's corner, and down through its floor. Run 1 comes through the living room ceiling in the alcove's right corner to Deco B; run 2 follows the right upright's groove down and through the floor to the kitchen, where Deco C goes on the outside of the brick cupboard, up high.
- **Sloped ceilings:** bedroom 2 and the bathroom slope down to the back wall; keep any Deco or desk in bedroom 2 on the high side.

What each part does

Part	Where	Job	Why it's needed
Virgin Hub 5, in modem mode	Bedroom 3	Brings the internet in, nothing else	Two routers fighting (double NAT) breaks Xbox online play; its Wi-Fi switches off
Deco A (main)	Bedroom 3	The router: Wi-Fi, parental controls, HomeShield	HomeShield and parental controls only work when the Deco is the router
Switch 1, 8-port	Bedroom 3	Turns Deco A's spare port into seven	The two PCs, both Cat6 runs down, and the optional projector and music-room runs
Deco B	Living room, high shelf at 196	Wi-Fi for the living room, and the wired point for the wall	Fixes the living room's weak 5 GHz side

Switch 2, 8-port	Standing on edge in the channel behind the base	Turns Deco B's spare port into seven	The Series X, Xbox One, Xbox 360, PS3, PC and TV each get a cable, one port spare
Deco C	Kitchen, high on the outside of the brick cupboard	Wi-Fi for the kitchen	The kitchen gets poor signal on every band today
Cat6 runs 1 and 2	Bedroom 3 → airing cupboard → alcove corner, and on to the kitchen	The wired link between Decos ("Ethernet backhaul")	Without it the Decos relay each other over the 5 GHz band the phones use, losing speed at each hop

Stage 1: the mesh up today, before any cables

1. Unplug everything from the hub. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its settings page moves to 192.168.100.1.
2. Connect a hub port to Deco A and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another port and restart the hub.
3. Plug Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Check the Xbox's network settings show NAT: Open.

Stage 2: after your Cat6 runs (step 3)

1. Plug run 1 into Deco B's free port (Switch 2 stays in the other) and run 2 into Deco C.
2. In the Deco app, check Deco B and Deco C show as "Ethernet".
3. Plug the Series X, Xbox One, Xbox 360, PS3, PC and TV into Switch 2.
4. Under QoS, give the Series X and the PC high priority; turn on guest Wi-Fi and automatic firmware.

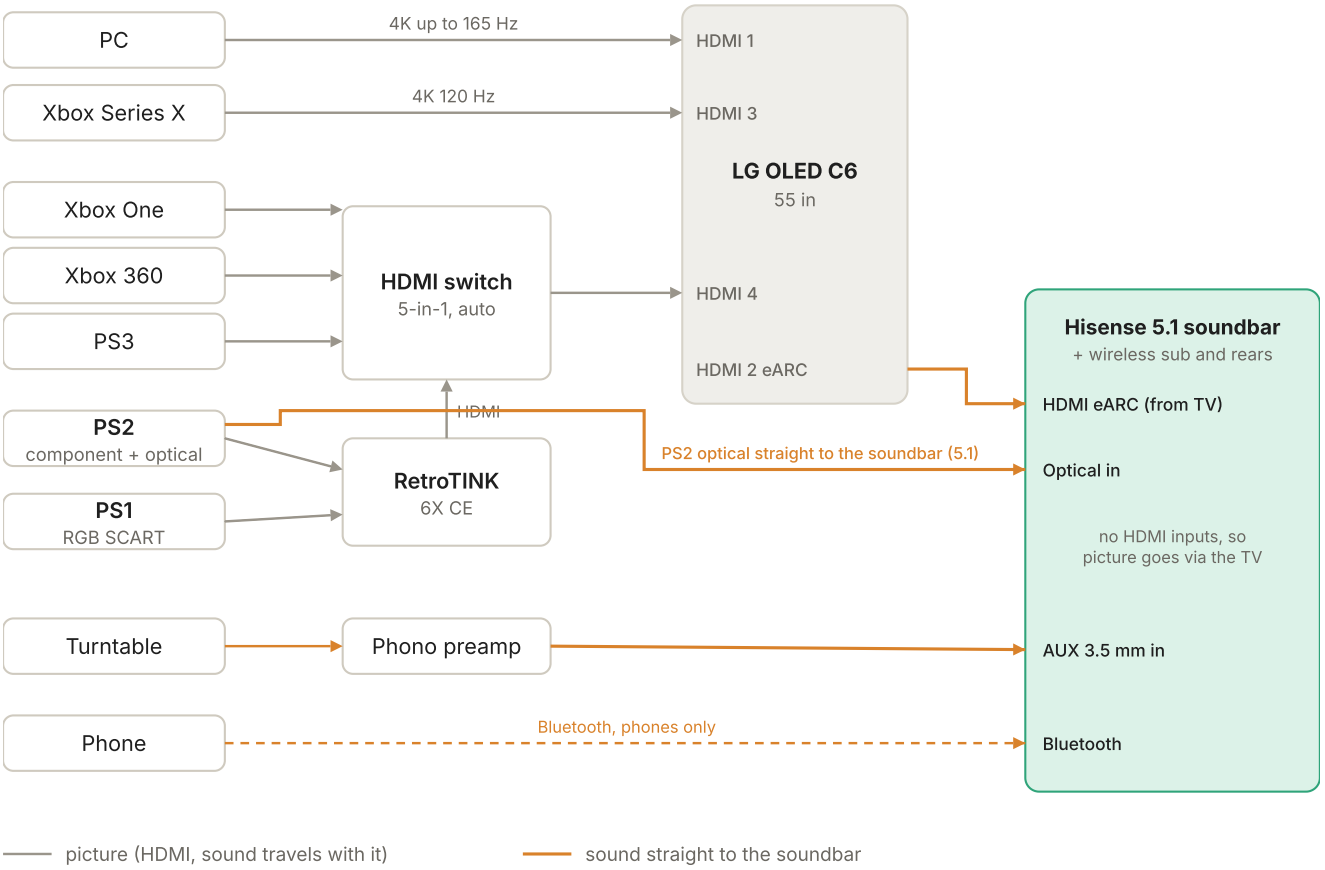
Cabling rules

- Solid pure-copper Cat6, round; not CCA or flat cable for the long runs. Measure the real route and add 2–3 m.
- Cross mains at right angles; never staple through a cable; no kinks; label both ends. Never two cables between the same two devices.
- **Done when:** Deco B and C show "Ethernet", each PC shows 1.0 Gbps, the Xbox shows NAT: Open, and phones hold a strong signal in the kitchen and across the living room.

AV: everything sounds through the Hisense

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



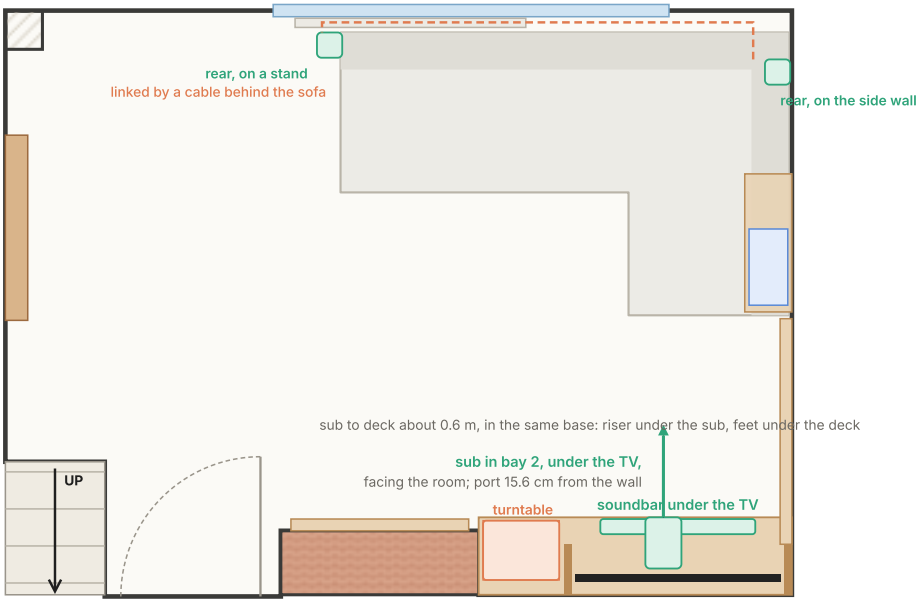
Source	Picture	Sound
PC · Xbox Series X	TV HDMI 1 · HDMI 3	Via the TV over eARC (5.1 or Atmos)
Xbox One, Xbox 360, PS3	HDMI switch → TV HDMI 4	Via the TV over eARC, 5.1
PS2	Component → RetroTINK → switch → TV	Optical straight to the soundbar
PS1	RGB SCART → RetroTINK → switch → TV	Stereo through the RetroTINK and eARC
Turntable · phone	—	Preamp → AUX · Bluetooth

TV settings: rename the inputs (HDMI 4 "Retro"), Deep Colour on HDMI 1, 3 and 4, Game Optimiser on the console inputs, sound out to HDMI eARC. RetroTINK-6X CE: a 5 V 2 A USB plug on the retro 6-way; the PS2's Component Video Out set to Y Cb/Pb Cr/Pr; RGB SCART for the PS1.

Sound layout

Sound: the sub on the front wall with the TV, the rears behind the sofa

To scale, north at the top. The sub is wireless: on the front wall it blends with the soundbar, out of the way of feet



- **Sub:** wireless, in bay 2 under the TV, its driver facing the room and its rear port 15.6 cm from the wall through the open back (15–30 is advised[25]). Next to the soundbar, its bass blends with the bar instead of coming from behind you, and it's off the floor where feet and drinks are. If the bass booms, turn it down rather than moving it.
- **Rears:** one on the side wall at seated ear height behind the sofa's corner, one on a stand past the sofa's west end; linked by a cable behind the sofa, powered from the socket by the sofa if there is one (if not, from a 12-way strip in place of the 10-gang).

10 · Build guide, for a first-time builder

This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Each step carries a coloured label for its part of the house (Prep, Network, Power & cables, Paint, Media wall, Dining table, Lighting, AV), and so does each phase of the buy list. You're doing the electrics yourselves: steps 3 and 12 have the safe routine and the point where you stop and call an electrician. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **The electrics are yours, within limits.** Swapping the socket for a double and the two ceiling fittings are like-for-like jobs, not notifiable under Part P in England, as long as they meet the wiring regulations. Isolate, lock off and prove dead every time. Anything new (a new circuit, the consumer unit, work by a bath or shower) is an electrician's, and so is anything on the stop list in section 9.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15
Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day

Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
ELECTRICS (YOUR OWN: STEPS 3 AND 12)			
Two-pole voltage tester to GS38, with its proving unit	proving every wire dead before you touch it: non-negotiable	Buy	35–70
MCB lock-off and a small padlock	keeping the circuit off while you work on it	Buy	8–15
Plug-in socket tester	checking the new double socket is wired the right way round	Buy	8–15
Insulated (VDE) screwdrivers, wire strippers, side cutters	the socket and the ceiling fittings	Buy	20–40
Wall chaser with dust extraction (or a bolster and club hammer)	the PC's two conduits in the side wall	Hire	25–45 a day
Cat6 punch-down tool and a cable tester	terminating and checking the Cat6 runs	Buy	20–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.
2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	PREP **Measure and count**	The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square.	Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.	2 h
2	NETWORK **Network, stage 1**	Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open.	Internet in every room, and NAT: Open on the Xbox.	2 h
3	POWER & CABLES NETWORK **Cables and the socket, your own job**	The Cat6: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner for Deco B, one on to the kitchen for Deco C, a spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Low voltage: no rules beyond care. The socket: swap the single on the side wall for a double at the same spot, with the circuit isolated, locked off and proved dead (the safe routine is in this step). Like for like, so not notifiable. The PC's two conduits: chase them into the solid side wall from behind the PC shelf to skirting height and behind the skirting to the base, 38 mm oval for the HDMI and Cat6, 25 mm for the PC's mains lead, a draw wire in each. Make good. Stop and call an electrician if anything on the stop list turns up (old wiring, no earth, a reading where there should be none, anything you can't match). Before you fill a chase, photograph it with a tape measure in shot, and mark every route on the plan.	The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed.	2–3 days
4	PAINT **Paint**	Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting, door, frame and radiator. Leave it a week to cure before fixing anything to it.	Two even coats, and no white showing at the edges or behind the radiator.	2–3 days, then a week to cure
5	PREP **Mark the heights**	With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds.	Every line level end to end, and the uprights' lines plumb.	1 h

6	MEDIA WALL **Build the base**	Order the oak and ply from the cutting list (below the steps), a week or two ahead: most merchants cut to size for a small charge. Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in.	The tops level within 1 mm across 186 cm, and nothing rocks.	1 day
7	MEDIA WALL **Fit the worktop**	Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move.	Tight to both walls, and the slot sits over the channel.	½ day
8	MEDIA WALL **Fit out the base**	Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (its cut list is with this step). Bay 3: the 12 mm divider first (17.8 cm in from the left side), then the LP crates' runners at 2.3 and 37.0 cm on the right and the three slim drawers' at 2.3, 25.1 and 47.9 on the left, then each box. The hidden plinth drawer under bay 2. Bay 2: the divider and the part shelf, then the sub on its riser, its front flush with the face frame.	Every drawer runs smoothly, the two crates clear each other by 2 cm, and the slim drawers by 2.8.	1–2 days
9	MEDIA WALL **Uprights and shelves**	On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings. On the chimney breast: the “now playing” ledge, its top at 104, on two concealed brackets, 6 cm in from each side of the breast (with the table by the stairs).	Uprights plumb, shelves level, the TV slides in with 2 cm each side, and the “now playing” ledge is level with an LP standing on it.	2–3 days
10	MEDIA WALL **The side-wall run and the PC end**	Cut the run from its 225 × 30 × 4 board: mark the front edge from the full-size template (7 cm deep along the LPs, sweeping out to 28 under the PC, a 5 cm radius on the end), jigsaw just outside the line and rout to it with a bearing-guided bit; rout the LED groove under the front edge. Screw the oak cleat along the side wall and fit the concealed brackets for the PC end. Lift the run on, dowel its end to the left upright's front edge, and screw it down to the cleat from above (plugged). Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the two chased conduits with the draw wires: data in the big one, its mains lead in the small one. The PC's extras (v17): screw the powered USB hub under the run's front edge below the PC, fit the remote power button's card and stick its button under the wrap ledge by the sofa, and lay the magnetic filter on the top exhaust.	One level line with the ledge over the TV, the sweep runs smooth with no flat spots, and the PC end holds your weight leaning on it.	1 day

11	DINING TABLE **The flip-top dining table**	Screw the backboard to the wall, plugs every 20 cm, and glue and screw the 7.3 cm hinge rail along its top. Fit the two Hebgo brackets on its face, 80 cm apart, arm tops level with the rail's top. On the bench: rout the three flip-top hinges into the meeting edges of the top and the half, and fit them; hinge the gate leg under the half, 3 cm in from its far edge, with its two stays. Screw the lift-off hinges to the rail's front edge and the top's back edge, all the same hand, and hang the pair. Fit the two turn buttons and the magnetic catch above it.	It opens onto both brackets with a click each side, the half flips out flush and its gate leg locks upright; it folds back flat and swings up to the wall without binding, and nothing moves when you lean on it open.	1–2 days
12	LIGHTING **Lighting: smart spot bars and the LED strips, your own job**	Isolate the lighting circuit at the consumer unit, lock it off and prove it dead at each ceiling point. Take down the two old fittings. The light switches stay as they are: the smart bulbs need full power, so no dimmers. Hang bar 1 on the rose between the TV and the sofa and bar 2 on the pendant point by the stairs, earth first, both running from the TV towards the window; fit the smart colour bulbs and point every head straight down, evenly spread; tilt one only if its pool lands badly. The LED strips in the uprights' grooves and under the side-wall shelf on their plug-in driver (on the 10-gang), the ceiling strip through its notches, the bias light on the TV; your red lamp in its spot. Power back on, switches on: add the eight bulbs to the app, group them as bar 1 and bar 2, and save the five lighting scenes (Paint and light), General first.	Every bulb answers the app, the scenes recall, and from the sofa no spot head shows in the screen or shines in your eyes.	1 day
13	AV NETWORK **Connect and finish**	TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.	Everything on the network's "done when" list, and every source plays through the soundbar.	1 day

Roughly 14–18 working days spread over a few weekends, the electrics and the Cat6 included, plus the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the LP crates before the anti-tip brackets go in.** A full crate weighs about 30 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

The cutting list

Every made part in one list, grouped by what it's cut from, for the timber merchant. Sizes are finished sizes in cm, length × width × thickness. The cut lists in the other sections say how each part joins; this list is for ordering.

Solid oak, hardwax oil

Ask for it planed all round; scribe the uprights and worktop on site

Part	Size (cm)	Qty	Notes	Section

Uprights	159 × 30 × 4.5	2	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench	Media wall
High shelf, long length	128.9 × 30 × 4	1	Housed 1.2 into both uprights and pegged	Media wall
High shelf, short length	51.7 × 30 × 4	1	Housed into the right upright; cleat on the reveal	Media wall
Series X shelf	55 × 30 × 4	1	Two through-tenons 6 × 2.4, wedged flush	Media wall
Record ledge	128.9 × 7 × 2.4	1	Housed and pegged	Media wall
Side-wall run (v15)	220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board	1	On an oak cleat, concealed brackets under the PC end, dowelled to the left upright; LED groove under the front edge	Media wall
Ledge lips	128.9 × 1.5 × 0.9 · 126 × 1.5 × 0.9	1 · 1	Glued along the ledge's and the run's front edges	Media wall
Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1	Screwed under the PC shelf	Media wall
Worktop	186 × 46 × 3	1	From one 300 × 62 full-stave worktop; chamfered front	Media wall

Dining backboard · hinge rail	110 × 22 × 3 · 110 × 4 × 7.3	1 · 1	From the same worktop's offcut	Dining table
Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1 set	End stiles scribed to the walls	Media wall
Table top and the half, centres	101 × 80 × 2.7	2	Edge-glued staves: 96 between the shoulders, plus a 2.5 tongue each end (v15: the half matches the top)	Dining table
Breadboard ends	80 × 7 × 2.7	4	Pegged with three 10 mm pegs each, in slotted holes	Dining table
Gate legs	73.3 × 4.5, tapering to 3	2	Tapered on the two inside faces; tenoned into the apron (v15)	Dining table
Gate apron	94 × 7 × 2	1	Between the legs; two butt hinges along its top edge, under the half	Dining table
Turn buttons	6 × 2.5 × 2.2	2	On a single screw each	Dining table
Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Finger pull cut underneath	Media wall
LP crate lips · slim drawer lips	36.3 × 5 × 1.5 · 16.5 × 5 × 1.5	2 · 3	Cut to the face frame's openings, less 3 mm (v16)	Media wall

“Now playing” ledge	105.5 × 7 × 2.4, lip 1.5	1	On the breast at 104, two concealed brackets (v16; with the table by the stairs)	Media wall
Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Offcuts, felted underneath	Media wall
Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6 + 6 · 8	Glued	Media wall

Birch ply, 18 mm

Part	Size (cm)	Qty	Notes	Section
Bay 3 sides	73 × 40	2	The record carcass, standing on the floor	Media wall
Bay 3 top and bottom	56.4 × 40	2	Between the sides; back left open to the channel	Media wall
Plinth rails	120 × 9	2	Front and back of the ladder frame under bays 1–2	Media wall
Plinth cross-pieces	34.4 × 9	3	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer	Media wall

Birch ply, 12 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer sides	35 × 15.5	2		Media wall
Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1		Media wall
LP crate bases · slim drawer bases	34.9 × 40 · 15.3 × 40	2 · 3	The sides and ends stand on them (v16)	Media wall
Bay 3 divider	69.4 × 38	1	17.8 in from the left side, the full height inside (v16)	Media wall
Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2	A 52 × 34 × 5 box	Media wall
Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1	The sub's column on the left; the PS2 over the PS1 on the right	Media wall
Keyboard tray	54 × 26	1	Oak lip on its front edge	Media wall

Birch ply, 9 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer base	52.5 × 34.4	1	In a 6 mm groove	Media wall
LP crate sides · ends	40 × 8 · 33.1 × 8	4 · 4	Two of each per crate: a 33 cm trough (v16)	Media wall

Birch ply, 6 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer cross divider	51.9 × 9	1	13 cm back from the front	Media wall
Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2	Floors raked so the cases lean about 13°	Media wall
Plinth drawer base	52 × 34	1	Screwed under the box	Media wall
Slim drawer sides · backs and fronts	40 × 10 · 14.1 × 10	6 · 6	Three boxes 15.3 wide, 14.1 inside (v16)	Media wall

Hardware

Runners are sold in pairs

Part	Size (cm)	Qty	Notes	Section
LP crate runners · slim drawer runners	400 mm full extension, 68–70 kg a pair · 400 mm light duty	2 pairs · 3 pairs	Into bay 3's sides and the divider (v16)	Media wall
Game drawer runners	350 mm full extension, soft-close	1 pair	Accuride DZ3832EC or Screwfix 964VN	Media wall
Plinth drawer and keyboard tray runners	300 mm full extension · 300 mm light duty	1 pair · 1 pair		Media wall
Häfele Hebgo folding brackets	287.43.428, 580 arm	2	500 kg a pair	Dining table
Lift-off hinges	Union PowerLoad 100 × 88, all the same hand	3		Dining table
Flip-top hinges	Lee Valley 40574, solid brass, 1.27 × 7.0	3	Routed into the meeting edges of the top and the half	Dining table
Gate hinges · folding stays	75 mm steel butt hinges · locking stays	2 · 2	The gate swings down and locks; folds flat inside the half	Dining table
Sub isolation riser	dense foam or sorbothane, about 22 × 28 × 3	1	Under the sub in bay 2	Media wall
Concealed brackets	for the uprights; two for the PC shelf, 40 kg+ a pair	1 set		Media wall

11 · Buy list and tracker

Every purchase and job, in build order, comes to about £3,110–6,337, with no electrician on top: the electrics are your own job, and the tools for them are on the list. Each phase carries its part's coloured label. Estimates are approximate UK prices; changed and new items are marked. Tick items off and write in what you paid.

Phase	Estimate (£)	Biggest item
NETWORK 1 Network, now	51–70	two 8-port switches
POWER & CABLES NETWORK 2 Cables and power	265–605	the electrical safety kit and Cat6
PAINT 3 Paint	220–540	paint
MEDIA WALL 4 Base unit	457–892	record trays and the BESTÅ frames
MEDIA WALL 5 Oak joinery	745–1,355	the solid oak worktop
DINING TABLE 6 Dining wall	576–1,130	the oak table top
LIGHTING 7 Lighting	288–795	the eight smart bulbs
AV 8 TV, consoles and AV	288–420	RetroTINK-6X CE
AV 9 Vinyl kit	80–190	phono preamp
AV NETWORK 10 Finishing	140–340	the rug
Total	**3,110–6,337**	**no electrician: the electrics are your own job (section 9)**

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" and its stand, a 10-gang, the Hisense AX5100G 5.1 soundbar, the Series X fan stand, the turntable, the L-sofa and all the consoles.

Status	Item	Qty	Estimate (£)	Paid (£)
NETWORK **1 NETWORK, NOW**				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
POWER & CABLES NETWORK **2 CABLES AND POWER**				
☐ To do	Cat6, your own job: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner (Deco B) and one on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2 **(changed: no electrician)**			
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	Cat6 tools: a punch-down tool, RJ45 keystones and two surface boxes (or plugs and a crimper), a cable tester, a draw-rod set **(new: you run the Cat6 yourselves)**	1 set	30–60	
☐ To do	Swap the single socket on the side wall for a double at the same spot: like for like, not notifiable; step 3 has the safe way to do it **(changed: your own job)**			
☐ To buy	A double socket in the same finish, and a deeper back box if the old one is shallow **(new)**	1	5–15	
☐ To buy	Electrical safety kit: a two-pole voltage tester to GS38 with its proving unit, an MCB lock-off and padlock, a plug-in socket tester **(new: for doing it yourselves)**	1 set	45–90	
☐ To buy	Insulated (VDE) screwdrivers, wire strippers and side cutters **(new)**	1 set	20–40	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To do	Chase two conduits into the solid side wall for the PC, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth: 38 mm oval for the HDMI and Cat6, 25 mm oval for the PC's mains lead; make good and paint **(changed: your own job; the PC now runs from the 10-gang)**			
☐ To buy	Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires, one oak-coloured brush grommet; a 5 m C13 lead with a rewireable 13 A plug for the PC **(changed)**	1	15–35	
☐ To buy	For the chase: a wall chaser with dust extraction for a day (or a bolster chisel and club hammer), one-coat filler **(new)**	1	20–45	
☐ To buy	PC tidy kit: braided sleeve, shelf-notch grommet	1	5–15	
☐ To buy	A powered USB 3 hub, 4-port with its own adapter, screwed under the side run's front edge below the PC: pads, the keyboard and a stick within reach of the sofa **(new in v17)**	1	15–30	
☐ To buy	A remote power button for the PC: a wireless switch on the motherboard's power header (card in a spare slot), its push button on the wrap ledge **(new in v17)**	1	10–30	
☐ To buy	A magnetic mesh dust filter cut to the PC's top exhaust **(new in v17)**	1	5–10	
☐ To buy	Only if there's no socket by the sofa: a 12-way surge-protected strip in place of the 10-gang, so the rear speakers get a way **(new)**	1	0–60	
PAINT **3 PAINT**				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door and frame; primer and undercoat		200–500	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
● MEDIA WALL	**4 BASE UNIT**			
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☑ Done	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)**	1	30–50	
☐ To buy	Bay 3 carcass: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts, and a 12 mm birch-ply divider 38 × 69.4 that splits it into the slim game column and the LP column **(changed in v16: the divider)**	1	50–90	
☐ To buy	Two face-out LP crates, 34.9 × 40 (a 33 cm trough): 12 and 9 mm birch ply cut to size, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg **(changed in v16: snug crates, no dead space)**	2	120–200	
☐ To buy	Three slim drawers for the game column, 15.3 × 40: 6 mm birch ply sides and backs, 12 mm bases, oak lips, felt, three pairs of 400 mm runners (light duty is enough); a 2 m USB-C lead and a grommet for the pads drawer **(new in v16)**	3	70–130	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads **(new in v13.5: the sub moves into the media wall)**	1	25–45	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
● MEDIA WALL	**5 OAK JOINERY**			
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, the side-wall run from one board 225 × 30 × 4 (cut to its sweep), lips **(changed in v15: one run for the ledge and the PC shelf)**	1 set	230–380	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)**	1	25–45	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
● DINING TABLE	**6 DINING WALL**			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Solid oak for the flip-top: two panels 110 × 80 × 2.7 with pegged breadboard ends, the top and the half (edge-glued staves from a timber merchant, who can also radius the corners) **(changed in v15: the half replaces the leaf)**	1 set	280–480	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Three Lee Valley 40574 narrow flip-top hinges (solid brass, 1.27 × 7.0 cm) for the joint **(new in v15)**	3	25–45	
☐ To buy	The gate leg: two tapered oak legs 73.3 × 4.5 to 3 and a 94 × 7 × 2 oak apron (from the cutting list), two 75 mm butt hinges, two folding locking stays **(new in v15: instead of eight legs and the leaf's leg case)**	1	40–90	
☐ To buy	Oak for the hinge rail (110 × 4 × 7.3), two oak turn buttons, a magnetic catch, felt pads **(changed in v15)**	1	15–30	
☐ To buy	Folding chairs, IKEA TERJE or similar, as many as you'll seat (skip if yours fit: seat about 45–48) **(new)**	4–7	100–280	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	
☐ To buy	The “now playing” ledge on the breast, with the table by the stairs: oak 105.5 × 7 × 2.4 with a 1.5 lip (from the shelf board's offcut), two concealed brackets **(new in v16)**	1	10–25	
☐ To do	Fix the backboard and hinge rail, hang the brackets and the top, fit the half on its flip-top hinges and the gate leg under it; check it opens, locks, folds and folds up			
LIGHTING	**7 LIGHTING**			
☐ To do	Swap both ceiling fittings for the spot bars, turned to run from the TV towards the window: like for like, your own job; step 12 has the safe way. The light switches stay as they are (smart bulbs need full power) **(changed in v15: no dimmers)**			
☐ To buy	Eight smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W (Wi-Fi, no hub), or Philips Hue White and Colour E14 lustre, 5.1 W, if a WiZ shows below its shade; both under the bar's 6 W **(new in v15)**	8	100–360	
☐ To buy	Smart colour strips for the upright grooves and under the side-wall run, in the same app as the bulbs: about 4.5 m of RGB and tunable-white strip (Philips Hue Lightstrip Plus, or WiZ's strip kits, cut at their marks), three aluminium channels with opal diffusers, plug-in power on the 10-gang (no fused spur); a 6500K bias light for the TV stays plain white **(changed in v16: smart colour, so the whole wall changes with a scene)**	1	90–220	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars, £49 each (bulbs above) **(changed: two)**	2	98–110	
☐ To buy	Optional: a smart button for the bars, so a scene is one press rather than the app (WiZ remote or Hue dimmer switch) **(new in v15)**	1	0–25	
☐ To do	Set up the smart bulbs in their app: group them as bar 1 and bar 2, then save the five scenes (General first; Paint and light) with their levels and colours; your red lamp's bulbs go in the same scenes **(new in v15)**			
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
AV	**8 TV, CONSOLES AND AV**			
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
AV **9 VINYL KIT**				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
AV NETWORK **10 FINISHING**				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			
☐ To buy	A smart plug with a schedule for the plant stand's grow bars (about 12 hours a day), and drip trays or a waterproof mat under the pots on its slatted tiers **(new in v18)**	1	15–30	
☐ To buy	Felt pads under the 2 × 2 record shelf, and one anti-tip strap to the breast (it holds about 160 LPs, about 35 kg) **(new in v18)**	1	5–10	
☐ To buy	A rug about 230 × 160 in front of the sofa: cream with deep blue, rust and ochre goes with the green, the oak and the blue sofa (the plan draws one as a guide) **(new in v15)**	1	120–300	

12 · Still to confirm, later ideas, sources

Still to confirm

- ☐ The stairs wall: solid, or plasterboard on studs? It decides the table's bracket fixings.
- ☐ The electrics: you're swapping the two light fittings and the socket yourselves (as planned; no dimmers since v15), or an electrician for the fittings?
- ☐ The sofa's return: how far it runs along the side wall (the long side is about 266 from your photo; the return is drawn at 168).
- ☐ Your new plant stand's width, depth and height (drawn at 90 × 30 × 120 from your photo).
- ☐ Your 2 × 2 record shelf's size and LPs per tray (drawn at KALLAX 2 × 2's 77 × 39 × 77, about 40 a tray).
- ☐ The stairs wall's length and your shelf's height (drawn at 196: the OBEGRÄNSAD needs 198 or lower to stand on it).
- ☐ The turntable's model (the zone takes up to 45.2 × 35.2 cm) and whether it has a PHONO/LINE switch.
- ☐ The Series X fan stand's model: the shelf takes a footprint up to about 30 cm wide.
- ☐ Dining chairs: use ones you have (seat 45–48), or folding ones (the flip-top seats up to six or seven)? And where they'll be kept.
- ☐ Records: roughly how many LPs (the wall holds about 191, all face-out)?
- ☐ Games: roughly how many PS2, Xbox 360 and Xbox One games? The drawers hold about 92.
- ☐ Wired ports for the projector (master) and the music room (bedroom 2)?
- ☐ The 10-gang: is it surge-protected, where are its two switches, and how long is it? It now feeds the PC too.
- ☐ Console model numbers, from the labels underneath, to tighten the row heights; the plan assumes the largest version of each.
- ☐ The pendant point by the stairs wall (the photo gives about 55 cm out, a quarter of the way along the shelf).
- ☐ The RetroTINK-6X CE's size when it arrives (bay 1 allows 23 × 23 × 8).

Easiest way to answer: the viewer page has a box for each of these and takes photos; or reply in a message, a line each.

Agreed so far

- ✓ From your v13.6 message: the legs store as part of the table (in the leaf's frame); the electrics are your own job; each step labelled with its part.
- ✓ From your v13 message: less cladding (done: no slats, spine, sub box or reeding; cables chased into the wall); the stairs go up; the drawer layouts drawn.
- ✓ From your answers (v12.1): the sofa's long side ends in line with the breast's reveal; the door hinges on the breast side and opens in; the side wall and breast are solid (plugs and screws, no stud fixings); the PC is an average gaming case (allowed for at 23 × 45.3 × 46.6).
- ✓ Artisan oak, Card Room Green with matching eggshell skirting, every record face-out, the game drawer as designed.
- ✓ The TV 25.25 cm nearer the side wall, the turntable in the alcove, the Series X above it, the records below.
- ✓ The ledge wrapping round onto the side wall, with the PC above the sofa.
- ✓ A fold-down oak dining table on the chimney breast; the spot bars on the existing ceiling points; the sub back in the media wall, in bay 2 (v13.5).

Later ideas

- **For the room (v17):** a glass display case on the breast, glass plant shelves in the window, a C-table for the sofa's open end, and small ones (v18: TV backlight that follows the picture, cushions in the rug's colours, a plant uplight, a rug pad): see the appendix “Ideas for later”, and the Ideas tab and buttons in the 3D room.
- **Dave and Emma's watch log:** a shared film and TV log, like a Spotify Blend for what you watch: log each film or episode, mark whether you watched it together, rate it, and later get suggestions for what to watch next. To be shaped up separately.

Sources

Numbered sources are the item sizes used in the fit proof (section 3).

1. Your hand-measured sketch of the living room, in mm (alcove 1860 × 380, chimney breast 1175, ceiling 2350)
2. LG: OLED55C6 specifications: 1222 × 703 × 45.1 mm without stand, 1222 × 757 × 230 mm with stand, stand 470 × 230 mm, 14.1/15.9 kg, VESA 300 × 200 (lg.com, OLED55C6 product pages)
3. Trusted Reviews and Expert Reviews, Hisense AX5100G reviews: soundbar 920 × 60 × 90 mm, subwoofer 214 × 294 × 304 mm (W × H × D) with a rear-firing port, rear speakers 90 × 144 × 109 mm
4. Wikipedia, Xbox Series X: 151 × 151 × 301 mm; Pure Xbox (Nov 2020) quoting Microsoft: at least 4 cm clear above the console on an open stand, 10–15 cm recommended on all sides
5. IKEA: BESTÅ frame 60 × 40 × 64 cm (10247379); BESTÅ shelf 56 × 36 cm, 20 kg maximum load
6. TP-Link: TL-SG108 specifications, 158 × 101 × 25 mm (tp-link.com)
7. Laptop Outlet: TP-Link Deco X1500 3-pack listing, 110 × 110 × 114 mm per unit
8. Audio-Technica: AT-LP120XUSB specifications, 452 × 352 × 141.6 mm, 8 kg without dust cover (audio-technica.com)
9. RetroTINK: RetroTINK-6X CE shop page (power 5 V 2 A by USB; no size published)
10. UGREEN: 5-in-1 4K60 HDMI switch CM568 (90512), 127 × 59 × 25 mm
11. Original Xbox One power supply: 17 × 7.5 × 4.8 cm (6.69 × 2.96 × 1.88 in), as listed by several replacement-PSU sellers
12. Wikipedia: Comparison of PlayStation consoles (PS1 270 × 188 × 60, PS2 302 × 183 × 78, PS3 325 × 274 × 98 mm); Xbox One (333 × 274 × 79 mm); Neowin: official Xbox 360 dimensions (309 × 258 × 83 mm)
13. Record and case sizes: LP sleeve 31.4 cm (12⅜ in); 7-inch sleeve 18.3–18.4 cm; DVD-style keep case 19 × 13.5 × 1.4 cm; Xbox One case 17 × 13.5 × 1.1–1.2 cm (videogamecasesize.info; Walvis Products; Wikipedia: Keep case)
14. Dunelm: Elements Wolston 4-light spotlight bar, £49: 84 × 16 cm, 22.4 cm drop, shades 15 cm, E14 6 W max (dunelm.com, product 1000189449)
15. Emuca heavy-duty full-extension runners, 400 mm, 68 kg per pair; Accuride DZ3832EC 350 mm, 43 kg; runner fitting allowance 12.5 mm a side
16. Xbox 360 original 203 W power supply: no Microsoft size published; Akyga's 203 W replacement for it (AK-PD-14) is 210 × 77 × 55 mm (b2b.akyga.com), so an allowance of 21 × 10 × 6 cm is used here (measure yours)
17. Turntable Lab: Audio-Technica replacement dust cover for the AT-LP120XUSB (705-140XP-1815), 17.375 × 13.625 × 2.75 in (44.1 × 34.6 × 7.0 cm)
18. Häfele UK, Technology catalogue 2023, p. 7.206: Hebgo heavy-duty folding bracket 287.43.428, arm 580 mm, height 220 mm, 500 kg per pair, locks automatically when raised; up to 800 mm between brackets (Häfele 287.43.419 listing)
19. Screwfix: Union PowerLoad lift-off hinges, 100 × 88 mm, 3-pack (5938T), 120 kg maximum door weight
20. Emuca: table legs Ø60 mm, 710 mm, height adjustable 710–730 mm, 50 kg per leg (set of 4): 710 is the standard dining leg; the plan's oak legs are made to the same 71 cm
21. RS PRO 45 mm steel spring clip (RS 175-3588): 42.1 × 56 × 12.7 mm, packs of 10 (uk.rs-online.com)
22. IKEA: TERJE folding chair, 43 W × 49 D × 78 H cm, seat height 48 cm, tested for 110 kg
23. Book a Builder UK, small dining room ideas: 600 mm of table edge per diner; table edge to a wall 600 mm tight minimum, 750 mm to aim for, 1000–1100 mm to walk behind; dining height about 750 mm
24. Your photo of the sofa (18 Oct 2026) and your note (21 Oct): a blue corduroy L-sofa, two seats then the corner on the long side, which ends about 40 cm short of the window's west end (about 266 cm long); the return's 168 is a typical corner-sofa size until measured
25. Klipsch subwoofer placement guidance, as quoted by Cloudspress and TechYorker (2026): leave about 6–12 in (15–30 cm) between a rear-firing port and the nearest obstruction
26. Xbox Support (Microsoft), About the Xbox One power supply (archived): don't block its vents, don't put it in a confined space such as a bookcase, rack or stereo cabinet unless the space is well ventilated, don't stack devices on it, and lay it flat on its four rubber pads; the Xbox 360 brick's advice is the same
27. Xbox Support (Microsoft), Xbox 360 placement tips, as quoted on Gaming Stack Exchange: an open, well-ventilated area, a hard surface, and away from high-vibrating sources such as subwoofers
28. Your two photos of the west end (20 Oct 2026): the 2 m black shelf on the wall beside the stairs, scaled from its 2 m: it starts about 41 cm past the stair box and stops about 16 cm short of the window corner, top about 200; the pendant hangs about 55 cm out from the wall, a quarter of the way along it
29. Corsair 4000D Airflow, a typical mid-tower gaming case: 453 × 230 × 466 mm (TechPowerUp review)
30. Lee Valley: 40574 narrow flip-top hinges, solid brass, 1/2 in × 2 3/4 in (1.27 × 7.0 cm), for card and flip-top tables (leevalley.com); their 40208 card-table hinge is 1 3/16 in (3.0 cm) wide, too wide for a 27 mm edge
31. Your photo (20 Oct) and note: Dunelm hessian pencil-pleat curtains; Dunelm pencil-pleat drops run 137, 182 and 228 cm with a 3 in header; Dunelm 28 mm wooden poles, 120–360 cm, rings included (one per 10 cm), finials up to 8 cm more (dunelm.com)
32. Your plant stand, from your photo (20 Oct): bamboo, two columns, four slatted tiers, LED grow bars under the top two. Drawn at 90 × 30 × 120 (W × D × H), the usual size of a two-column four-tier bamboo stand (e.g. VIVOSUN's 4-tier bamboo stand, 47.2 in = 120 cm tall: homedepot.com); please measure yours

33. IKEA: KALLAX shelving unit 2 × 2, 77 × 39 × 77 cm, cubes 33 × 33 inside (ikea.com). Your 2 × 2 wooden record shelf with four trays is drawn at this size until you measure it
34. IKEA: OBEGRÄNSAD LED wall lamp, black (your 905.262.63; 005.262.48 and others by country): 52 × 36 × 6 cm, 5 W, five lighting programmes, hangs landscape or portrait, runs on a 5 V USB-A adapter sold separately (ikea.com); its face is a 16 × 16 grid of white LEDs. From the Swedish House Mafia collection (IKEA press kit, Sept 2022)
35. Sony: DualSense controller 160 × 66 × 106 mm (W × H × D); Microsoft: Xbox Wireless Controller 153 × 61 × 104 mm
36. Record-shop browser crates and LP storage boxes are made about 32.5–33 cm inside for 31.4 cm sleeves (31.8–32.4 in outer sleeves), flipped by the sleeves' top edges
37. Philips Hue Lightstrip Plus V4: 2 m base kit, cuttable every 33 cm, extendable with 1 m; WiZ Wi-Fi LED strip, 2 m starter kit, cuttable; both RGB and tunable white, in the same apps as the bars' bulbs (philips-hue.com; wizconnected.com)
38. Smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W, 4.5 × 10.1 cm, Wi-Fi, no hub (Philips UK specification sheet, 8719514437333); Philips Hue White and Colour Ambiance E14 lustre, 5.1 W, 45 × 77 mm (Philips Hue; Lampmasters)
39. Virgin Media: Hub 5 modem mode
40. RetroTINK: Introducing the RetroTINK-6X CE (inputs including TOSLINK, deinterlacing); RetroTINK-2X Pro listed as legacy; RetroTINK-4K CE
41. Expert Reviews: LG C6 OLED review (4 × HDMI 2.1, one eARC, ports on the left); JB Hi-Fi: LG C6 specifications (165 Hz mode for PC sources)
42. Hisense AX5100G spec sheet (inputs, auto power-on); JB Hi-Fi Business: AX5100G (rears linked by cable, wireless to the soundbar)
43. TP-Link: Deco X1500 specifications (two gigabit ports per unit)
44. Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the tray capacity
45. Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
46. NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
47. Sony PS3 and Microsoft Xbox 360 S manuals: ventilation; IKEA UK: secure it
48. Ideal Home: colour drenching; The Paint Shed: colour drenching with Farrow & Ball; Livingetc: sage green shades (Card Room Green, Treron, Sage Green, Green Smoke)
49. Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours)
50. Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
51. haart Medway: floor plans and listing photos for the house (Portsmouth Close)
52. ConsoleMods: PS2 Component Video Out setting
53. GOV.UK: Electrical safety, Approved Document P (2013 edition, for use in England): notifiable work is a new circuit, a replacement consumer unit, or an addition or alteration to a circuit in a special location (round a bath or shower); like-for-like replacements and additions to existing circuits elsewhere are not notifiable (gov.uk/government/publications/electrical-safety-approved-document-p)
54. IET Electrical: Part P, England and Wales (notifiable and non-notifiable work); Planning Portal: Electrics, Building Regulations (non-notifiable work must still meet the regulations, BS 7671)

Appendix · Ideas for later (not in the plan or the costs)

Suggestions only, for later: they aren't in the plan, the buy list or the costs (all of them together would be about £370–725). The first three can be seen in the 3D room with the “Ideas (not in the plan)” buttons, all off until you turn them on.

Idea	Why, and where	About £
A slim glass display case on the chimney breast 90 wide × 50 tall × 12 deep	For the hats or collectibles, lit from inside, without a cabinet on the floor: 12 deep, so it stands no further out than a picture ledge. Oak frame from the offcuts, a 4 mm toughened glass front that lifts off, an LED strip under the top on the strips' app. It takes the place of the two framed LPs, above the “now playing” ledge. *On the breast, centred, from 137 to 187 (just clear of the sleeves on the “now playing” ledge); with the table by the stairs only.*	60–120
Glass plant shelves across the window reveal two 10 mm toughened shelves, 235 × 18, at 130 and 170	Your deep sill already takes a row of plants; two glass shelves above it make the window a green screen that still lets the light through. Studs into the reveal's sides, and a slim steel rod down from the head at the middle so a 235 span can't sag. *Inside the window's 22 cm reveal, in front of the glass, clear of the curtains.*	120–200
An oak C-shaped side table for the sofa's open end top 45 × 30, 60 tall	The ottoman end has no arm to put a drink on. A C-table's top slides over the end of the seat and its foot sits flat on the floor beside it, so drinks and pads are at hand and it moves away when someone lies along the seat. Made from the oak offcuts. *At the sofa's open end, towards the TV.*	25–60
TV backlight that follows the picture a sync box on the HDMI line and a strip behind the TV (in place of the plain bias light)	The wall behind the TV glows with the colours on screen, in the same app as the bars and strips, so a film can tint the room. *Behind the TV, on the bias light's place; the box sits by the HDMI switch in bay 1.*	60–130
Cushions and a throw in the rug's rust and ochre two or three 45 × 45 cushions and a throw	They tie the blue corduroy sofa to the rug and the oak, and soften the ottoman end. *The sofa.*	40–90
A small uplight under the plant stand's bottom tier a plug-in spot, about 10 cm	Throws the leaves' shadows up the wall at night, so the plants are part of the evening lighting even with their grow bars off. *On the floor, under the stand.*	10–20
A non-slip pad under the rug 220 × 150, trimmed to fit	Keeps the rug flat and square on the carpet where the sofa's open end and the coffee table push it. *Under the rug.*	15–30
A smart scene button by the door a small wireless button, about 5 × 5	One press for the General scene coming in, a long press for everything off, in the same app as the bars' bulbs and the strips. *On the wall beside the door's light switch.*	15–25
A motion nightlight on the stairs battery LED strips under the treads' noses, or a single motion light	A soft glow on the stairs at night with nothing to wire; it turns itself off. *The stairs.*	15–30

A record-cleaning tray in the pads drawer a small tray, about 30 × 15	The carbon brush, a stylus brush and a cleaning fluid kept together next to the deck. *Bay 3's pads drawer.*	10–20

Idea only, from the 3D room: glass plant shelves across the window's reveal, beside your plant stand in the west corner.

Idea only: the slim glass display case on the chimney breast, lit inside, above the “now playing” ledge.

Appendix · Earlier rounds

v15. Your notes on the room as it really is: **your sofa** is drawn in blue corduroy, two seats then the corner, its long side running to about 40 cm short of the window's end, with no cushion; **your red lamp** with its colour bulbs (green on top, pink, then purple); on your shelf by the stairs, **your IKEA light box** at the left end, **your laser** in the middle (its dots on the ceiling) and the **plant hanging** down off the shelf; **your hat** on an oak peg by the stairs; **soft grey carpet** (your photo shows it already is), a **designed rug** that fits in front of the sofa (230 × 160), and **your curtains** from your photo: long hessian pencil-pleat panels to the floor on a wooden pole just under the ceiling strip (open or drawn in the 3D room). The **record trays take LPs only**: one trough the full width, with room to flip with both hands. The **side wall** looks better: the ledge and the PC shelf become **one oak run** that sweeps out in a curve to the PC, with a warm LED glow underneath. **Both spot bars now run from the TV towards the window** and take **smart colour bulbs** (E14, under the bar's 6 W), so there are no dimmers to fit; the light is set softer and aimed at walls and shelves, for less spotty light. In the 3D room every light has a **slider**, the bars and the lamp a **colour**, and the **walls, ceiling and skirting** each have their own paint colour. And a **new dining table that stays on the wall and seats more**: a flip-top gate-leg table. Its second half flips out on brass hinges to make 110 × 160, on a folding oak gate leg: four with room, six elbows-in, seven by the stairs; no lifting off, no leaf or legs to store (section 5). Your **curtains** are Dunelm hessian on a wooden pole with rings and ball finials.

v13. You asked for less cladding all over the place, so it's gone: no slatted cable spine, no slatted PC screen, no slat panel behind the turntable, no slatted sub box, no reeded table. The PC's leads now run in conduit **chased into the solid side wall**, so there's no trunking either; the sub stands under a **plain oak cover** (below); the folded table is a **plain oiled oak panel** between its pegged breadboard ends. The oak that's left is the joinery that does a job: the uprights, the shelves, the ledges, the worktop, the face frame and the table. The **stairs go up** to the first floor (drawn rising away from the room; tell us if they run the other way). New: **the drawer layouts**, every drawer to scale with what goes in it (section 3). The slat-wall alternative is retired.

v13.1–13.2: the sub gets a **plain oak cover** with no sides: spills run off it, the sub can't tip, and nothing blocks its driver or port (section 5). A new render shows **every drawer out**, and the viewer can lay **rough measurements** over any view. The **stairwell opening** now starts just after the door, over the banister, and the sofa is drawn as yours. **Six at the table** now use the leaf too: 170 × 80, 85 cm a diner on the long sides and 75 cm off the breast, every clearance checked. The **side view** now shows how deep the alcove is round the wall: 38 of the 46 cm base inside it, 8 proud, the uprights and shelves wholly inside.

v13.3: the build guide is now step by step, each step with what to buy before it and what to check after it; the buy list shows what each phase costs; and the network and power section lists what each part does, both setup stages and what plugs into every socket (section 9).

v13.5: the sub is back in the media wall. The two Xbox bricks lie flat at the back of their shelves by the power leads, the RetroTINK and the HDMI switch move into the space that frees in bay 1, and the sub takes the lower left of bay 2, under the TV, on an isolation riser beside the PS2 and PS1 (section 3). The dining wall has nothing under it now, so eight can sit down without moving anything.

v13.7: four **lighting scenes** (TV night, dinner, records, everyday evening), each drawn on the room plan with every light's level, so you know where to set each dimmer (section 8).

v13.8: the whole plan is back on your own deck (the one you can edit, with your saved answers), and the summaries now follow the fit proof and the questions list, so they can't fall out of date: the PC case is answered and off the list, and the three checks on site are the TV's plugs, the RetroTINK's size and the sub's vibration.

v13.6: the **leaf carries the legs**: a slim oak frame underneath holds all eight oak legs in clips, so the folded table stays a plain panel, nothing new stands in the room, and the leaf goes wherever the chairs are kept (section 5). **The electrics are your own job**: swapping the socket for a double, the two dimmers and the two spot bars are like-for-like and not notifiable in England, so there's no electrician's visit; there is a safe routine and a clear point to stop and call one (section 9). The PC now runs from the 10-gang. Every step and every buy-list phase carries a **coloured label** for its part (Network, Media wall, Dining table and so on), and one list says what we still need from you.

v14: a **second place for the table**: on the wall by the stairs, under your 2 m shelf (section 5 compares the two). The plan page now has a **3D room** you can walk round: switch the table's wall, fold it or set it for four or six, change the paint, the lighting scene and where the floor lamp stands, and save any view as an HD picture.

Appendix · Options we set aside

Options we rejected along the way

Option	Why not
Slatted oak cladding: cable spine, PC screen, turntable panel, sub box, reeded table (v12)	You asked for less cladding: the chased conduit and plain oak do the same jobs more quietly
The slat wall alternative (v7–v12)	Cladding over the whole alcove; the oak uprights won
The vinyl cabinet on the chimney breast (v10)	The breast is the only clear wall for a table; the turntable fits in the alcove now
Hub as router, Decos as access points	Loses HomeShield and parental controls
All-wireless mesh · powerline	Speed drops at each hop · slower and less stable than Cat6
Records spine-up in one drawer	Showed only a handful of covers; face-out trays show every one
A green feature wall only · white skirting	The drench ties the room together; white cuts a bright line round it
Bluetooth from the turntable	Compresses the sound; the AUX lead keeps full quality
Lifting the table off onto eight legs, with a 60 cm leaf that stores the legs (v13.1–v14)	You wanted it to stay on the wall and seat more: the flip-top's half and gate leg do both (v15)
Folding legs fixed under the table · a box for the legs under the dining wall · the plinth drawer (v13.6)	They'd show on the folded panel and stand it 5 cm off the wall · it sits where the chairs push back with six · too short for 71 cm legs. The leaf carries them instead.
An electrician's visit for the socket, dimmers and fittings (v13.6)	You're doing them yourselves: like-for-like, not notifiable; a new socket behind the sofa and a fused spur are dropped